FastSpeech 2 for English and Telugu

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Technical report: implementation, data, training and analysis of two separately trained FastSpeech 2 models (English: Svarah; Telugu: IndicTTS Telugu) · 7 November 2026

Code, results and checkpoints: `https://github.com/ashdev-7/iiith-fastspeech2`

How to read this report. All numbers were measured in this project unless they are marked *[paper]* (reported in the FastSpeech 2 paper [1]) or *[reference]* (taken from the reference implementation [2] or its configuration files). Statements marked *[interpretation]* are explanations that were not tested. Each measurement was produced by a scripted Kaggle job that records the code version it ran; the jobs are listed in `kaggle/README.md`, and the files each number comes from are named in the text.

Intelligibility was measured with automatic speech recognition (Whisper large-v3 [8] for both languages; for Telugu, transcripts that Whisper wrote in Kannada script are also scored after mapping them back to Telugu, Section 5.5), with the recognisers pinned to exact versions. No listening test was run, so the report makes no claim about perceived naturalness. Sections 5.1 to 5.3 and 7, and the extra sentences in Sections 7.3 and 7.4, go beyond what the assignment asks.

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Results at a glance

	English (Svarah)	Telugu (IndicTTS)
Training data used: hours (utterances)	4.87 h (3,077)	4.90 h (2,140)
Validation data used: hours (utterances)	0.98 h (742)	1.01 h (446)
Speakers in training (Section 2.1)	110 profiles	2
Training steps	17,949	55,524
Training time and hardware	211 min, 1 T4 GPU	480 min, 2 T4 GPUs
Validation loss, total: lowest (step) and last	2.651 (7k), 2.686	1.970 (16k), 1.974
Duration error on validation (frames) and log-duration correlation	2.81, 0.58	1.06, 0.85
Pitch error on validation (Hz) and correlation, per phone	34.6, 0.35	29.2, 0.76
Character error rate on validation, synthesis vs copy-synthesis [a]	15.7 % vs 11.6 %	35.8 % vs 33.1 %
Character error rate on the five test sentences [b]	3.6 %	27.1 %
Real-time factor, T4 GPU and 4-thread CPU	0.027, 0.35	0.026, 0.37

[a] Whisper large-v3 [8] on 150 validation utterances; copy-synthesis is the real recording's mel-spectrogram passed through the same vocoder, i.e. the best this vocoder and recogniser allow. Whisper is weak on Telugu (Section 7). [b] English: final configuration, after Whisper's English text normaliser. Telugu: after mapping Kannada-script output back to Telugu; much of the remaining error is recogniser formatting, Latin script in T1 and digits in T4 (Section 5.5).

All required outputs are in the repository: ten generated test sentences, thirty variance-adaptor plots, and training- and validation-loss plots for both languages (checklist in the Appendix). Every test sentence passes the run-time checks: the predicted durations sum to the mel length, no phone has zero length, and the audio is finite and not silent.

Beyond the brief, the report contains an evaluation on every held-out utterance (Section 7); a diagnosis of where quality is lost and three remedies (Sections 5.1 to 5.3); a check that the Telugu aligner's access to the validation clips does not bias the results (Section 2.2); 34 further sentences of our own, not from the assignment (Sections 7.3 and 7.4); and twelve fixes to the reference code (Section 1.3).

1. Implementation approach

1.1 Architecture used from the reference implementation

The starting point is `ming024/FastSpeech2` [2] (MIT licence), an implementation of FastSpeech 2 [1], vendored with the modifications in Section 1.3. The model is the non-autoregressive FastSpeech 2 acoustic model:

- **Encoder**: phoneme embedding + 4 feed-forward Transformer (FFT) blocks (self-attention, 2 heads, hidden 256; Conv1D 1024, kernels 9/1).
- **Speaker embedding** (multi-speaker mode) added to the encoder output: 110 Svarah training speaker profiles (Svarah has no speaker ids, Section 2.1), 2 Telugu speakers.

- **Variance adaptor**: three predictors (2 × Conv1D 256, kernel 3, LayerNorm, dropout 0.5) for log-duration, pitch and energy. Pitch and energy are *per phone*, quantised into 256 bins, embedded and added to the hidden sequence; the **length regulator** repeats each phone vector d times, with d = round(exp(log $\hat{d}$) − 1) at inference. In training the ground-truth duration, pitch and energy are used here (teacher forcing) and the predictors learn them; at inference the predictions are used.
- **Decoder**: 6 FFT blocks → linear to 80 mel bins → 5-layer convolutional **PostNet** (residual).
- **Loss**: MAE on mel (before and after PostNet) + MSE on log-duration, pitch and energy (6 terms logged).
- **Vocoder**: the reference repository's pretrained **HiFi-GAN "universal"** generator [3] (not trained here).

	FastSpeech 2 paper [1]	**Reference code [2]**	**This project**
Training data	LJSpeech, 24 h, 1 speaker	configs for LJSpeech, AISHELL-3, LibriTTS	about 5 h per language
Pitch target	frame-level F0, continuous wavelet transform	F0 averaged per phone, no wavelet transform	as the reference code
Decoder layers	4	6	6
Dropout	0.1	0.2	0.2
Batch size	48	16	16
Training steps	160k	900k (LJSpeech config)	17,949 (English), 55,524 (Telugu)
Vocoder	Parallel WaveGAN	HiFi-GAN or MelGAN	HiFi-GAN "universal" [3]
Listening test (MOS)	3.83 ± 0.08	not reported	not run
Real-time factor, with vocoder	1.95×10^{-2} (V100 GPU)	not reported	2.6 to 2.7 × 10^{-2} (T4 GPU)

The paper's figures cannot be compared directly with ours: the data (24 h of one speaker against about 5 h), the vocoder and the GPU all differ. Phone-averaged pitch is also used by FastPitch [10].

1.2 Language-specific work

Telugu (the reference has no Telugu support)

- **Grapheme-to-phone** (`text/telugu/g2p.py`, `phones.py`): Telugu script is close to phonemic, so it is converted by deterministic rules on the Unicode Telugu block: a consonant carries the inherent vowel /a/ unless a vowel sign or virama follows; a vowel sign replaces it; the anusvara (ం) becomes the nasal homorganic with the next consonant (బలంగా → `b a l a ng g aa`, ఉంచండి → `u ny c a nn dd i`), otherwise /m/; visarga → /h/. 50 phone labels in ASCII (retroflex consonants doubled: `tt dd nn ss ll`; aspirates with `h`; long vowels doubled). Every code point must be consumed by a rule, otherwise an error is raised, so characters are never dropped silently.
- **Normalisation** (`text/telugu/normalize.py`): Telugu number words for cardinals, ordinals and times: 2026 → రెండు వేల ఇరవై ఆరు, 12వ → పన్నెండవ, 10:45 → పది నలభై ఐదు.
- **Symbol table**: a separate 65-symbol Telugu table (padding, punctuation, 50 phones, silences); the encoder embedding is sized per language (L1–L4 in `MODIFICATIONS.md`, which also has a one-table summary of all Telugu work).

- **Alignment**: we found no pretrained Telugu acoustic model for MFA, so one was **trained here** on the selected IndicTTS clips with a dictionary generated by the G2P (11,947 words, 0 without pronunciation); Section 2.2 shows that its access to the validation clips does not affect the results.
- **Code-mixing** (beyond the test set): English words written in Latin script inside Telugu text (e.g. "iPhone") are mapped from ARPAbet to the nearest Telugu phones, using only phones already in the Telugu table (`text/telugu/latin.py`).
- **Plots**: Telugu sentences in plot titles are drawn with proper text shaping (Pillow with libraqm on Kaggle; for the final greyscale figures in this report, a headless browser via `tools/replot.py`, redrawn from the saved numbers), so conjuncts and vowel signs are not broken.

English

- **Inference frontend** (`text/english_frontend.py`): times ("10:45 AM" → "ten forty five A M", AM/PM as letter names), numbers and years (reference `inflect` rules: 2026 → "twenty twenty six"); homographs known to g2p_en (*conduct*) resolved by part of speech over the whole sentence; pronunciations from the LibriSpeech lexicon [14] (the lexicon used for alignment); g2p_en [13] for out-of-vocabulary words; punctuation → short pause `sp`. Every decision is recorded per word in `frontend_trace.json`; the test text itself is never changed.
- **Rules found by inspecting the test outputs** (Section 5.4), all general, none word-specific: split words ("rhyth m") are joined for pronunciation when the concatenation is a lexicon word and the first part is not; a word-final "oo" in an out-of-vocabulary word is /uː/ (*andaroo*); the **citation form** is preferred where the lexicon's first variant is a casual reduction (dropped /t/ in "twenty", "parliamentary", dropped /d/ in "thousand"; flapped /t/ in "thirty"; 670 words in all); "the" before a vowel is /ðiː/; and an out-of-vocabulary word of 12 or more letters is split into the fewest lexicon words of at least 4 letters (*antidisestablishmentarianism* → anti + disestablishment + arianism), instead of being left to g2p_en.
- **Robustness rules for unseen text** (from probing with new sentences; none changes the ten test sentences, checked phone by phone): "%" and "&" spoken; an hour with a.m./p.m. and no minutes; acronyms spelled or taken from CMUdict; "1,247" read as a quantity, not a year; "Month day" dates as ordinals; no pause after "Dr."; two corrections to g2p_en's homograph table (past-tense *read*, verb *process*).
- **Alignment**: MFA's pretrained `english_us_arpa` acoustic model with the LibriSpeech lexicon + g2p_en for out-of-vocabulary words (6,388 words, 0 without pronunciation).
- **Inference-time options** (Sections 5.2–5.3): a global-variance post-filter and clear-speech options, each introduced for a measured problem; the trained weights are the same for every output.

1.3 Modifications to the reference code

Twelve fixes (M1–M12) and four language-support changes (L1–L4) are listed in `MODIFICATIONS.md` with the original behaviour, the reason, the new behaviour and the test that covers it. The ones that change results:

#	Problem in the reference	Fix
M1	Energy embedding received the *pitch* control value	uses `e_control`
M5	Utterances without a TextGrid added the previous utterance's pitch/energy to the normalisation statistics	statistics only from processed utterances
M6	Empty TextGrid intervals (MFA ≥ 2 writes pauses this way) were dropped, so pauses vanished from phone sequences and durations	kept as `sp`

M9	float64 energy targets: `backward()` failed ("Found dtype Double")	cast to float32
M10	numpy-scalar learning rate in checkpoints, rejected by `torch.load(weights_only=True)`	`float()`
M12	Pitch/energy controls multiplied the *z-normalised* value: for a voice below the corpus mean, pitch × 1.2 *lowered* F0 (126 → 113 Hz)	scale in Hz / energy units; now × 1.2 → 126 → 157 Hz (en, former voice, commit a2c8afe), 209 → 252 Hz (te)

Others: CPU-safe STFT (M2), librosa ≥ 0.10 keyword API (M3), UTF-8 file handling (M4, M7), sorted speaker ids (M7), `map_location`/`weights_only` loading and HiFi-GAN paths (M8), masks on the input's device (M11).

1.4 Engineering for a reliable run

All compute ran as Kaggle script jobs that clone the repository and record the commit. Every job except the data-ingest and data-accounting jobs first runs the test suite (64 tests at the Telugu training job, 108 in the last jobs: frontends, symbol tables, split leakage, preprocessing, model forward/backward on every test sentence, checkpoint round trip, plots, prosody and clarity rules) and stops if a test fails; a CPU **rehearsal** ran every stage on a small real subset before the full runs. The training job measures throughput for 300 steps, sizes the schedule to a time budget, and runs a **dry run of inference and plotting** on that 300-step checkpoint before the long training. A **watchdog** kills a training process that exceeds its budget or stops logging; it was added after the first English run hung inside training (two-GPU DataParallel) and was killed at Kaggle's 12-hour limit with no checkpoint. The rerun used one GPU and completed.

1.5 What was inherited, what we changed, what we added

Part	Origin	Our change	Evidence
Model (encoder, variance adaptor, length regulator, decoder, PostNet), loss, optimizer	ming024/FastSpeech2 @ d4e79eb	none to the architecture; encoder vocabulary sized per language (L3); control fixes M1, M12; device fixes M11	diff vs upstream: `model/fastspeech2.py`, `model/loss.py`, `transformer/Layers.py` unchanged
Hyper-parameters	reference LJSpeech configs	max_seq_len 1000 → 1500 (clips up to 15 s); steps and LR decays sized to a time budget	`English/config.yaml`, `Telugu/config.yaml`
Vocoder	reference: pretrained HiFi-GAN "universal"	none (a fine-tune was tried and not adopted, Section 5.2)	`hifigan/` unchanged
English text → phones	reference: LibriSpeech lexicon, g2p_en, English cleaners, number rules	new inference frontend with traced decisions and general rules (Section 1.2)	`text/english_frontend.py`, tests
Telugu text → phones	nothing in the reference	new: normaliser, rule-based G2P, 65-symbol table, Latin-script mapping	`text/telugu/`, `text/symbols.py`, tests

Data loaders, split	reference: LJSpeech/AISHELL-3/LibriTTS loaders, unseeded random split	new HF ingest, candidate filter, group split with leakage checks	`data/ingest.py`, `data/select_split.py`
Alignment	reference: MFA run externally	English: MFA pretrained `english_us_arpa`; Telugu: MFA acoustic model trained here	`data/mfa_prepare.py`, `kaggle/pipeline.py`
Feature extraction	reference preprocessor	bug fixes M5–M7; manifest-driven split	`preprocessor/preprocessor.py`
Inference, plots, evaluation, post-processing	reference: `synthesize.py` only	new (inference with checks, variance plots, evaluation, GV post-filter, prosody and clarity options)	`inference.py`, `plotting.py`, `evaluate_val.py`, …
Data, checkpoints	datasets named by the assignment; no checkpoints supplied	both models trained here from random initialisation	job records `fs2-03-train-*`

1.6 Data flow and tensor shapes (from the code)

Shapes recorded with forward hooks on both models (batch 1; L phones including pauses, T mel frames):

Stage	Component	Output
text → phones	English frontend / Telugu normaliser + G2P (language-specific)	phone list + per-word trace
phones → ids	`text_to_sequence` (table per language; unknown symbol raises)	$(1, L)$ int64
embedding + encoder	`Encoder`: embedding, sinusoidal positions, 4 FFT blocks	$(1, L, 256)$
speaker	`speaker_emb`, added to every position	$(1, 256)$ → broadcast
variance predictors	duration (log), pitch, energy, one value per phone	$(1, L)$ each
pitch / energy embedding	256-bin quantisation → `nn.Embedding(256, 256)`, added	$(1, L, 256)$
length regulator	repeat phone i d_i times, $T = \Sigma d_i$	$(1, T, 256)$
decoder → mel	6 FFT blocks, Linear$(256, 80)$, PostNet (residual)	$(1, T, 80)$ log-mel
vocoder	HiFi-GAN universal, input $(1, 80, T)$	$256 \cdot T$ samples at 22,050 Hz

In training (and in the validation loss of Section 4) the ground-truth duration, pitch and energy are embedded and used by the length regulator; the predictors are trained against them. At inference the predictions are used. Everything below the frontend and the symbol table is language-independent code, trained separately per language; the two models share no weights.

2. Data: selection, split and preprocessing

2.1 Selection and subsampling

Both datasets are larger than the 5 h + 1 h budget, so a deterministic, duration-targeted selection was used (`data/select_split.py`). Only the two specified datasets are used, Svarah [4] and IndicTTS Telugu [5], from Hugging Face at pinned revisions (`ai4bharat/Svarah`, `SPRINGLab/IndicTTS_Telugu`).

- **Candidates** (`data/ingest.py`): clips of 1–15 s with a non-empty transcript and non-silent audio. Svarah (Hugging Face split "test", 6,656 clips): 1,476 clips outside 1–15 s were excluded, leaving 5,180 candidates (7.97 h). IndicTTS Telugu (8,576 clips): 3,201 outside 1–15 s and 47 whose transcript the Telugu G2P cannot convert (mostly a Latin "o" typed in place of the anusvara) were excluded, leaving 5,328 candidates (11.84 h; female 3.77 h, male 8.07 h). No duplicate audio was found. Excluded transcripts were logged, not repaired (Kaggle `fs2-01-ingest`, `Evaluation/Data/job_records/`).
- **Svarah speakers are a proxy.** Svarah has no speaker-id column; its file names identify recordings, not speakers. A "speaker" here is a metadata profile (gender, age group, primary language, native state and district): 110 distinct profiles, while the dataset card states 117 speakers, so some speakers presumably share a profile and are merged under one embedding *[interpretation]*.
- **Unit of assignment = group**, never a single clip: a Svarah *recording* (all chunks of one recording), or for Telugu all utterances with the same normalised transcript (both IndicTTS speakers often read the same sentence). Groups that share a transcript are additionally merged (union-find), so no sentence can occur in both splits.
- **Order**: groups are visited in a seeded pseudo-random order (SHA-1 of seed and group id), independent of file order.
- **Quotas**: for Svarah each speaker profile gets train/val hours proportional to its share of the candidate audio; for Telugu the two speakers get equal quotas (`--balance equal`), because their candidate audio differs (3.77 h vs 8.07 h); a group goes to validation if it fits all validation quotas it touches, else to training if it fits, else it is not used.
- **No overlap**: utterance ids, groups and normalised transcripts must be disjoint between train and validation, and clips with identical audio (SHA-1 of the PCM samples) are excluded before the split. All checks passed (0 overlaps).

	English (Svarah)	Telugu (IndicTTS)
Selected train	3,079 utt. · 4.87 h · 110 speaker profiles	2,164 utt. · 4.96 h (female 2.41 h, male 2.55 h)
Selected validation	742 utt. · 0.98 h · 109 speaker profiles	450 utt. · 1.02 h (0.51 h each)
Not aligned by MFA	0	28 (1.1 %), mostly in runs of near-consecutive ids
Rejected in preprocessing	2 (≤ 1 voiced frame)	0
Used for training / validation	3,077 / 742	2,140 / 446
Hours used, train / validation (selected audio)	4.87 / 0.98 h	4.90 / 1.01 h
Same, after leading/trailing silence trimming (mel frames)	4.53 / 0.92 h	4.21 / 0.87 h

Overlap of the used sets: ids / transcripts	0 / 0	0 / 0

Hours and overlaps of the used sets: Kaggle `fs2-13-data-accounting`, `Evaluation/Data/accounting.json`. The unaligned Telugu clips are mostly in runs of near-consecutive ids (te_00915–00929: 6, te_01062–01072: 3, te_02330–02333: 4, te_02638–02654: 11; the other 4 lie between te_00111 and te_00191), which suggests transcript/audio mismatches in those ranges rather than random failures *[interpretation]*; they were excluded, not repaired. Svarah validation speaker profiles also appear in training (different recordings), and both Telugu speakers are in both splits: the split is by recording / sentence, not by speaker, so the validation sets measure generalisation to new sentences, not to new voices.

2.2 Preprocessing

1. **Audio**: decoded from the Hugging Face parquet files, mono, resampled to 22,050 Hz (soxr), 16-bit.
2. **Text**: English transcripts through the reference `english_cleaners` (lower case, numbers and abbreviations spelled out), keeping letters and apostrophes; pronunciations from the LibriSpeech lexicon with g2p_en for missing words. Telugu transcripts through the Telugu normaliser; pronunciations from the Telugu G2P. The inference frontends use the same lexicon, G2P and phone sets.
3. **Forced alignment** (Montreal Forced Aligner [6], MFA 3.4.3): English 3,821 / 3,821 utterances aligned; Telugu 2,586 / 2,614. The English aligner is MFA's pretrained model and saw none of our data. The Telugu aligner had to be trained here; it was trained on the selected training and validation clips (no other audio), because the aligner also produces the validation duration targets. The FastSpeech 2 model itself trains only on the training split. To measure whether the aligner's access to the validation clips matters, a second Telugu aligner was trained on the **training clips only** (same dictionary, same MFA version and command), the validation clips were aligned with it, and both alignments were compared (`tools/aligner_check.py`, Kaggle `fs2-18-aligner-check`, `Evaluation/Data/aligner_check.json`):

Second aligner (training clips only) vs the one used	Validation clips (445)	Training clips (2,139)
Phones whose duration is within 1 frame (11.6 ms) / identical	95.4 % / 66.8 %	96.1 % / 69.1 %
Mean phone-boundary shift (95th percentile)	2.7 ms (10 ms)	2.6 ms (10 ms)
Clips with the same phone sequence (the others differ only in where a short pause was placed)	363	1,951

   The released model's validation duration error against the two alignments: **1.058 → 1.112 frames** (MAE) and Pearson r of log durations **0.848 → 0.823**. The reported Telugu duration scores are therefore optimistic by about 0.05 frames (0.6 ms per phone), and the training targets the model learned from would have been almost the same without the validation clips (agreement on the training clips is as high as on the validation clips).
4. **Features** (reference preprocessor with fixes M5–M7): leading/trailing silence trimmed using the alignment; 80-band log-mel (FFT 1024, hop 256 = 11.6 ms, 0–8 kHz, matching the HiFi-GAN vocoder); F0 with WORLD [7] DIO + StoneMask, unvoiced frames linearly interpolated, averaged per phone; energy = L2 norm of each STFT frame, averaged per phone; pitch and energy z-normalised with the training-set mean/std; per-phone durations in frames from the TextGrids.

3. Training configuration

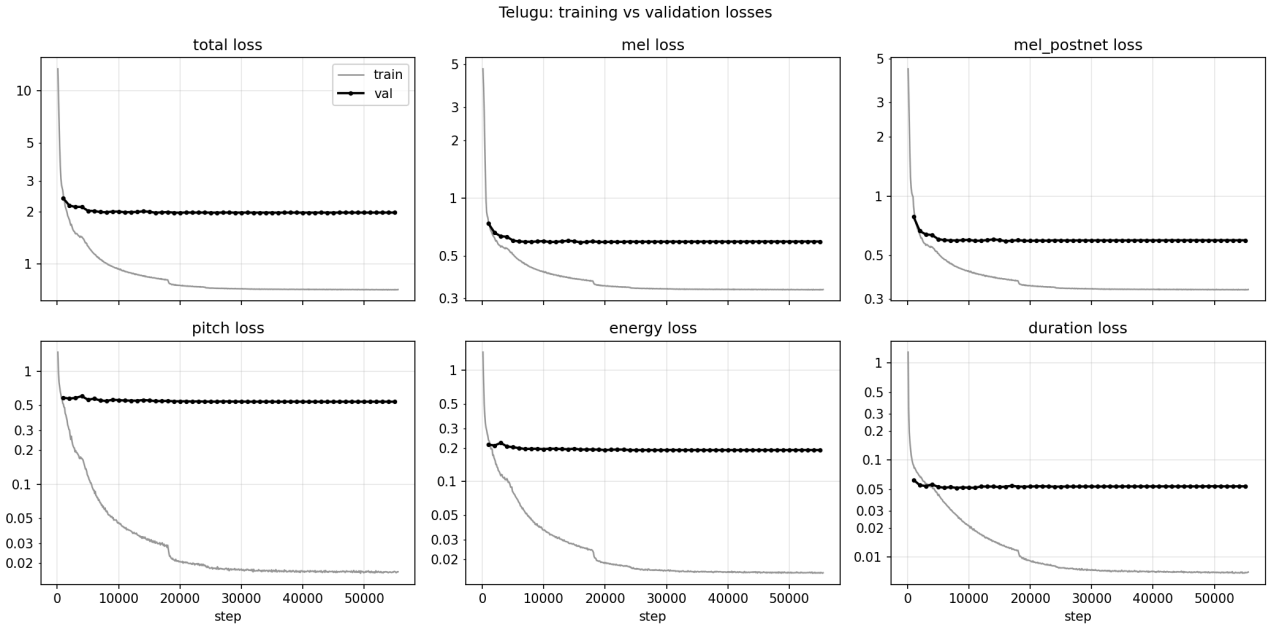
Both models use the reference architecture and optimiser settings (Section 1.1); only the length of training differs.

Setting	English	Telugu
Trainable parameters	35.44 M	35.34 M
Symbol table / speaker table size	360 / 110	65 / 2
Batch size	16	16
Planned steps	20,000	56,000
Learning-rate decays (× 0.3)	6k, 8k, 11k	18k, 24k, 31k
Throughput in the 300-step probe	1.45 steps/s	2.08 steps/s
Time budget	210 min	480 min
Steps completed (stopped by the budget)	17,949	55,524
Training time	211 min	480 min
Hardware	1 T4 GPU	2 T4 GPUs

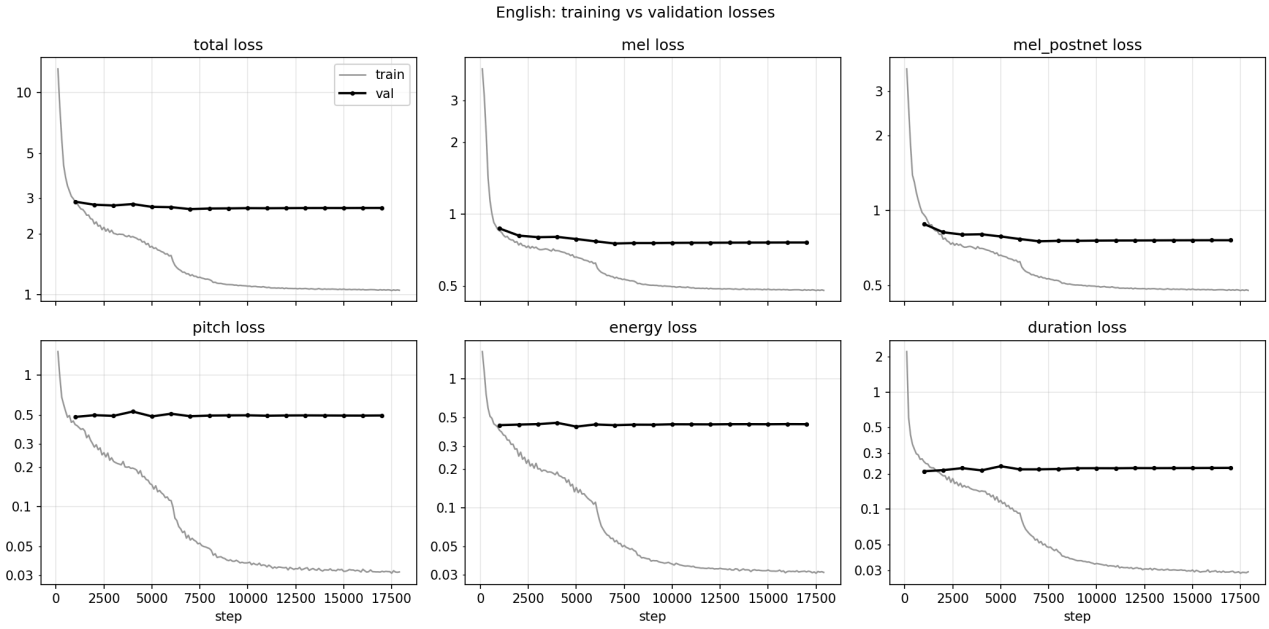
Settings common to both: Adam with $\beta = (0.9, 0.98)$ and $\varepsilon = 10^{-9}$; the warm-up schedule of [16] over 4,000 steps; learning-rate decays at 1/3, 4/9 and 5/9 of the planned steps, the same proportions as the reference's 300k, 400k and 500k of 900k; gradient clipping at 1.0; length-sorted groups of four batches; 32-bit floating point; random seed 1234. Validation runs every 1,000 steps on the full validation set and records the six losses, the duration error and the total-length error (`val_diagnostics.csv`). The parameter difference comes from the symbol embedding (75,520 parameters) and the speaker embedding (27,648).

The English budget was cut from 8 h to 3.5 h for the rerun because the Telugu validation total had come within 2 % of its final value by step 7k (Section 4); this also saved GPU quota and shortened the exposure to another session failure. Exact configs: `English/config.yaml`, `Telugu/config.yaml`, and as run: `Results/<Lang>/train_config_used.yaml`.

4. Loss curves: convergence, stability and over-fitting



Telugu: training loss (grey line, mean of each 100 steps) and validation loss (black line with dots, every 1,000 steps), logarithmic scale. Separate `training_loss.png` and `validation_loss.png` are in `Results/Telugu/`.



English, same layout (`Results/English/`).

Validation loss	English lowest (step)	English last	Telugu lowest (step)	Telugu last
total	2.651 (7k)	2.686	1.970 (16k)	1.974
mel after PostNet	0.753	0.760	0.589	0.595
pitch (z-units^2)	0.483	0.495	0.537	0.538
energy	0.427	0.445	0.192	0.192
log-duration	0.211	0.226	0.052	0.054

final *training* total	1.049	0.707

What the validation loss measures. It is computed like the training loss: the ground-truth duration, pitch and energy of each validation utterance drive the length regulator and the embeddings (`evaluate.py`). The mel terms therefore measure reconstruction with the true prosody, and the duration, pitch and energy terms measure the predictors on unseen phones. Free-running quality is measured separately in Section 7.

Convergence. The total training loss falls from 13 to ≈ 2 in the first 2k of the 4k warm-up steps, then slowly. The learning-rate decays show as steps: the mean training loss of the 1,000 steps after the first decay is 18.7 % (English, 6k) and 5.6 % (Telugu, 18k) below that of the 1,000 steps before, against 7.3 % and 2.2 % in comparison windows without a decay; the later decays change it by 6.2 % or less. The validation total is within 2 % of its final value from step 5k (English) and 7k (Telugu) onwards.

Stability. No logged training loss exceeds 1.3× the median of its neighbours (no spikes), no divergence, no non-finite losses (the trainer stops on any non-finite value; both runs stopped at their time budget).

Over-fitting. The gap between training and validation grows throughout: by the end (mean of the last 10 training logs) the training pitch loss is ~32× lower than validation for Telugu and ~15× for English, the energy loss ~13× and ~14×. The variance predictors show by far the largest train/validation gap (the mel terms 1.6–1.8×). For Telugu it is benign: the validation total ends within 0.2 % of its best; only the duration term ends 3.7 % above its minimum (8k). For English there is **mild over-fitting**: validation pitch, energy and duration losses end 2.6–6.9 % above their minima (reached at 1k–5k steps), and the total 1.3 % above its minimum (7k). The last validation is at 17k (English) and 55k (Telugu); the released checkpoints (17,949 and 55,524 steps) were not validated separately. Early stopping on the validation total (minimum at 7k steps) would give a slightly better English model and save time; the released checkpoint is the last one *[interpretation]*.

English vs Telugu level. English validation losses are higher in every term except pitch. The two runs differ in data (110 speaker profiles of ≈ 2.7 min each with varying recording conditions, Section 5.1, against two speakers with ~2.5 h each), frontend and training length (17,949 vs 55,524 steps), so the difference cannot be attributed to the language *[interpretation]*.

5. Generated speech, per test sentence

Audio: `Results/<Lang>/test_generated_audios/`. Voices: English `svarah_8a8e2655` (chosen by recording clarity, Section 5.1); Telugu `te_female` (lower HiFi-GAN copy-synthesis error than `te_male`: mel L1 0.378 vs 0.394 on 20 validation clips each). Pronunciation is assessed by transcribing each wav with an ASR model and comparing it with the frontend's normalised text (character error rate, CER); for English the CER is also given after Whisper's English text normaliser, which removes pure formatting differences ("10:45" vs "ten forty five").

Test text, as provided. The ten sentences are taken character for character from the text layer of the assignment PDF (PyMuPDF), which matches the rendered page; they are stored with their Unicode code points and SHA-256 in `tests/test_sentences.json`, and every inference run refuses a sentence whose hash differs. Other text extractions of the same PDF (pypdf, the Word copy) detach or reorder Telugu vowel signs (e.g. "విశ్లే ించారు" for "విశ్లేషించారు"), so they were not used. Spacing in the given text is kept as is, including "rhyth m" (E3) and "అడ విలో" (T3); the frontends only decide how such text is pronounced, and `frontend_trace.json` records the original text next to the model input. For "rhyth m" the English frontend

joins the pieces (read literally, "m" becomes the letter name: the first full run said "rythe em", Section 5.4); for "అడ విలో" no rule is needed, because the Telugu input has no word-boundary symbol and the split gives exactly the phones of అడవిలో.

5.1 Choice of the English test voice, and where quality is lost

The first English outputs (voice `svarah_71f88d2a`, chosen because it has the most training audio) sounded muffled. A Praat analysis [12] (`tools/acoustic_analysis.py`) and a diagnostic job (`fastspeech2/diagnose_quality.py`, Kaggle `fs2-07-diagnose`, `Evaluation/Diagnosis/`) located the cause. For 24–30 validation clips of each test voice it compares the natural recording, copy-synthesis (ground-truth mel → HiFi-GAN), FastSpeech 2 with ground-truth duration/pitch/energy (oracle) and free-running FastSpeech 2, band by band (long-term spectrum, dB relative to the natural recording):

Band (Hz)	0–500	500–1k	1k–2k	2k–4k	4k–6k	6k–8k
English, copy − natural	−0.1	+0.7	+0.5	+0.8	+1.0	+1.1
English, free-running − natural	−0.3	+3.1	0.0	+3.1	+3.9	+1.9
Telugu, copy − natural	0.0	+0.4	−0.7	+0.1	+0.2	0.0
Telugu, free-running − natural	+0.3	−0.3	−2.4	−1.9	−1.1	−0.2

On these clips (the first English voice and `te_female`) neither the vocoder nor the acoustic model removes high-frequency energy: both reproduce the recording to within ~1 dB (copy) and ~4 dB (free-running) in every band, and HNR, jitter and shimmer of all four versions are within 1.8 dB / 0.4 % / 1.2 % of each other (above 8 kHz, outside the mel range, the recordings have almost no energy and all synthetic versions add a low floor; not shown). The muffled sound was **the voice's own recordings**: ranking all 110 training speaker profiles by the clarity of their recordings (1–4 kHz band share + HNR/2, three training clips each, `clarity_en.json`) put `svarah_71f88d2a` 109th of 110 (1–4 kHz band −23 dB); recording conditions vary by speaker. The test sentences were rendered in the four clearest voices; `svarah_8a8e2655` (rank 2, 1–4 kHz band −9.7 dB) had the lowest jitter and shimmer of the four and is used for all English outputs. The checkpoint is unchanged; only the speaker embedding chosen at inference differs. Effect at the time (means over the five English sentences, Praat): spectral centre of gravity 298 → 623 Hz, jitter 1.18 → 0.51 %, shimmer 11.1 → 8.9 % (`Evaluation/Diagnosis/voice_choice.json`). The selection rule was changed after listening to the first outputs; it is an objective rule on training recordings, and the earlier outputs remain in the repository history (commit `8f5ae5f`).

5.2 Over-smoothing: diagnosis and three remedies

The band-energy comparison above cannot see whether spectral *detail* is preserved. Measuring the variance of each log-mel bin over time ("global variance", GV) and the mean frame-to-frame change (flux), on 6 of the held-out clips per language (the ones whose audio the job kept; English: the first voice), shows over-smoothing, a known weakness of FastSpeech 2 (its decoder is trained with an L1 loss on the mel here, and regression losses are known to average out spectral detail; we measured the reduced variance, not its mechanism). English free-running mels keep 80–88 % of the natural GV (three mel-band groups) and 80 % of its flux, and with ground-truth duration, pitch and energy 73–91 % and 79 % (so it is the decoder, not the variance predictors); copy-synthesis keeps 97–101 %. Telugu free-running keeps 93–98 % and 93 % of the flux (`Evaluation/Diagnosis/gv_bands.json`; a 6-clip measurement). A plausible reason, not tested, is that

with 110 speaker profiles of ~2.7 min each the English decoder averages over very different voices and recording conditions *[interpretation]*.

Remedy 1: GV post-filter (adopted for English). `fastspeech2/postfilter.py` rescales each mel bin's deviation from its utterance mean by $r_k^{s/2}$, where r_k is the ratio of natural to predicted variance, estimated once on **training** clips of the test voice (`Results/English/gv_ratio.json`), as in statistical parametric synthesis (global-variance compensation [9]). Inference only. Ratio estimated on 29 training clips (Telugu voice: 40); scored on the test voice's 9 held-out clips (Telugu: 25) (`Evaluation/Postfilter/`, Kaggle `fs2-08-postfilter`):

English, strength s	GV / natural	flux / natural	HNR (dB)	shimmer (%)	MCD (DTW)
0 (off)	0.76	0.62	16.9	8.2	36.7
0.5	0.90	0.67	17.4	8.1	37.6
1.0 (used)	1.04	0.71	17.6	7.9	38.6
Telugu 0 → 1.0	0.97 → 1.00	0.81 → 0.81	18.0 → 17.9	8.0 → 7.6	29.7 → 29.9

The post-filter restores the variance completely and part of the temporal sharpness; HNR rises by 0.7 dB and shimmer falls by 0.4 points (9 clips, small changes); MCD rises slightly because it rewards averaged spectra (a known property of MCD). Telugu is not over-smoothed ($r \approx 1.02$, no measurable change), so its outputs are left unfiltered.

Remedy 2: HiFi-GAN fine-tuning on ground-truth-aligned mels (tried, not adopted). The vocoder was fine-tuned per language on FastSpeech 2's teacher-forced mels of the training set, paired with the recordings cut exactly as the preprocessor cuts them, with discriminators (MPD + MSD, from the original HiFi-GAN code [3]) trained from scratch because the reference ships only the generator (`fastspeech2/vocoder_finetune.py`, Kaggle `fs2-09-vocoder-ft-*`; `Evaluation/VocoderFinetune/summary.json`).

Universal → fine-tuned	val. mel L1 (GTA)	Whisper CER, synthesis	Whisper CER, copy-synthesis	other
English	0.866 → 0.850	15.7 → 15.3 %	11.6 → 12.3 %	GV ratio 0.91 → 0.87; copy-synthesis HNR 14.9 dB vs 17.1 natural
Telugu	0.671 → 0.600	35.8 → 35.5 %	33.1 → 31.0 %	GV ratio 1.00 → 0.94

The fine-tuned vocoders fit the training mels better but did not make the synthesis clearer and made some outputs worse, so the universal HiFi-GAN is kept for both languages. The likely reason is the short adversarial training with discriminators started from scratch (fine-tuning normally starts from trained discriminators and runs far longer) *[interpretation]*.

Remedy 3: speaker adaptation of the decoder (tried, not adopted). The trained model was fine-tuned on the test voice's 29 **training** clips only (encoder frozen), with the step chosen on its 9 validation clips (`fastspeech2/adapt_speaker.py`, Kaggle `fs2-11-adapt-en`, `Evaluation/Adaptation/`). The training mel loss fell from 0.47 to 0.38, but the held-out clips improved only up to step 100–150 and then got worse; the gains (≤ 0.5 MCD on 9 clips) are within noise and the voice-quality measures do not move: with 2.7 min of one voice, the decoder fits the training clips without improving on the held-out ones. The unadapted model is kept.

5.3 Clear-speech inference options (English)

Listening to the English outputs exposed three further problems. Each was measured, and each remedy acts at inference time (`fastspeech2/clarity.py`, `prosody_rules.py`); the model weights are unchanged. The model's own predictions are kept in every variance file and drawn in every plot (Section 6), so nothing here hides what the variance adaptor predicted.

- **Endings cut off.** Training clips are trimmed at the first and last phone, so the model's last frame is still at speech level and the waveform stops mid-sound (last 20 ms of the model-only outputs: -42 dBFS). Remedy: a short silence lead-in, a 5-frame fade of the last frame and trailing silence before the vocoder (-86 dBFS).
- **Background noise.** The vocoder alone is clean (−86 dBFS RMS on a silent mel), but the model reproduces the Svarah recordings' noise floor in its pauses (43.8 dB above digital silence, measured on 119 pause frames of the model's own output on three neutral sentences that are not test sentences, `Results/English/noise_profile.json`). Remedy: subtraction of that floor in the mel domain. Quietest frames -46.6 → -54.2 dBFS; speech level -17.1 → -16.5 dBFS (unchanged) (`tools/clarity_measure.py` on E1–E4, model-only vs final; `Evaluation/English/clarity_levels.json`).
- **Fast, run-together speech.** On validation the model is 4 % shorter than the real speakers (median length ratio 0.96), and long words sounded run together with their neighbours (the author's listening impression, not measured). Remedy: overall duration × 1.10; words with four or more vowels × 1.25, preceded by a 70 ms micro-pause (placed before their article, never between article and noun); phrasing (Section 6: lengthening before punctuation, a minimum pause at each punctuation mark, a falling end for statements and a rising end for yes/no questions).

Which combination. Every configuration was rendered on Kaggle and scored with Whisper large-v3 on the five test sentences and 18 further sentences (Sections 7.3–7.4: 8 stress and 10 extended; XE01 is E1 verbatim, so E1 counts twice in "all 23"); CER after Whisper's English normaliser; the five test sentences total 335 reference characters, so one character is 0.3 points. The table is `Evaluation/English/variants_summary.json`:

English configuration (same checkpoint and voice)	CER, 5 test sentences	CER, all 23	E4
model + GV post-filter †	2.7 %	2.9 %	2.9 %
+ terminal fall †	2.7 %	3.2 %	2.9 %
+ fall + lexical stress	6.9 %	4.1 %	17.1 %
+ fall + stress, denoise, edge fade	7.8 %	4.0 %	15.7 %
final: fall + stress + phrasing, duration × 1.10, long words × 1.25 with micro-pauses, denoise, edge fade	**3.6 %**	**2.3 %**	17.1 %
final without the lexical-stress rule (Kaggle `fs2-21-final-en-nostress`)	3.9 %	2.3 %	15.7 %

† Earlier frontend (before the citation forms, the "the" rule, the compound split and the acronym rules), so these two rows are not strictly comparable with the others. Kaggle jobs: `fs2-12-prosody-en`, `fs2-14-extended`, `fs2-16-clarity`, `fs2-17-final-en`, `fs2-21-final-en-nostress`.

The final configuration has the lowest error over all 23 sentences (differences of under one point are a few characters and not significant), and the author preferred it on listening over the earlier configurations. Removing the stress rule changes little (2.28 → 2.34 %); the rule is kept because it puts the homograph contrast of E2a/E2b into pitch (Section 6.1). One sentence is worse than with the plain model: the last clause of E4 ("so andaroo should arrive on time") is garbled for the ASR in every configuration with the current frontend (15.7–17.1 %), with or without the stress rule, whereas the model with only the post-filter and the earlier frontend gave "So, Antaro should arrive on time" (2.9 %). The cause was not isolated; the stretch is a made-up word followed by function words that the model predicts very short (Section 6.1, E4).

Model-only renders of the five English test sentences (no inference-time option and no post-filter; the same frontend) are in `Evaluation/English/model_only/` for comparison.

5.4 English

ID	Duration	Whisper large-v3 transcript of the generated audio	CER, raw	CER, normalised
E1	8.65 s	The anti-disestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	0.0 %	0.0 %
E2a	5.53 s	The orchestra's conduct during the ceremony was praised by the audience.	0.0 %	0.0 %
E2b	6.01 s	The researchers will conduct an experiment to evaluate the proposed method.	0.0 %	0.0 %
E3	4.16 s	The rhythm of the orchestra changed unexpectedly.	0.0 %	0.0 %
E4	8.39 s	The meeting will start at 10.45 a.m. on 12 November 2026. Suhant Arashudra I. One time.	47.3 %	17.1 %

Frontend decisions (recorded in `frontend_trace.json`; the test text itself is unchanged):

- **E1**: "The" before a vowel → `DH IY0`; *antidisestablishmentarianism* is not in the lexicon → split into lexicon words anti + disestablishment + arianism (`AE1 N T AY0 D IH0 S IH0 S T AE1 B L IH0 SH M AH0 N T AE1 R IY0 AH0 N IH2 Z AH0 M`); *parliamentary* in its citation form (restored /t/); micro-pauses before *considerable* and *the parliamentary*.
- **E2a**: *conduct* tagged noun (NN) → `K AA1 N D AH0 K T`; micro-pause before *the ceremony*.
- **E2b**: *conduct* tagged verb (VB) → `K AA0 N D AH1 K T`; micro-pauses before *an experiment* and *to evaluate*.
- **E3**: "rhyth m" joined for pronunciation → *rhythm* `R IH1 DH AH0 M` (text unchanged); micro-pause before *unexpectedly*.
- **E4**: "ten forty five A M on twelve November twenty twenty six"; AM as letter names (durations × 1.6); *twenty* in its citation form (restored /t/); *andaroo* (Telugu అందరూ, "everyone") → `AE0 N D AA1 R UW0`.

Observations. Four of the five sentences are transcribed exactly (E4's raw CER is mostly digit formatting: "10.45", "12", "2026"). **E1**: the long rare word is recognised as the intended word; it needed the compound split and the slower long-word timing. **E2a/E2b**: the homograph pair is transcribed correctly, with the stress contrast in pitch (Section 6.1). **E3**: exact, including the joined *rhythm*. **E4**: the numbers, the date and *AM* are recognised; the last clause is not ("Suhant Arashudra I. One time."): *andaroo* is a Telugu word (అందరూ, "everyone") that the ASR cannot know, and the function words after it are short (Section 5.3). Across the five sentences the clear-speech options make the endings complete and the pauses quieter; on

listening (the author's impression; no listening test was run) short unstressed function words and a slightly breathy quality remain (Section 5.2).

Found by inspecting the outputs. The first full English run (same checkpoint) read E3 as `R AY1 DH` + the letter name `EH1 M` ("rythe em") and ended *andaroo* in `OW`. Both were caught in `frontend_trace.json` and fixed by general rules, covered by tests (Section 1.2); the outputs were regenerated with the unchanged checkpoint.

5.5 Telugu

ID	Duration	Whisper large-v3 transcript of the generated audio	CER	CER, script-mapped
T1	5.27 s	Neuro-Pharmacological పరిసోధనలో నూతన ఉషధాల ప్రభావాన్ని విస్లేశించారు.	38.7 %	38.7 %
T2a	2.46 s	తాడు పయతని పట్టు చాలా బలంగా ఉంది.	7.1 %	7.1 %
T2b	2.69 s	ఆమేపట్టు చిరను ధరించి వేడుకకు వచ్చింది.	5.9 %	5.9 %
T3	2.14 s	పులి అడవిలో నిస్సబ్దంగా నడిచింది.	6.9 %	6.9 %
T4	9.13 s	2026 నవం బర్పన్ నండవతేది ఉదయం 10.45 గంటలకు ప్రారంబమోతుంది సమయానికి సిద్ధంగా ఉంచండి	38.8 %	38.8 %

Frontend decisions (recorded in `frontend_trace.json`):

- **T1**: all words by rule; `ny` before `c` in విశ్లేషించారు.
- **T2a**: పట్టు → `p a tt tt u`.
- **T2b**: పట్టు → `p a tt tt u` (same as T2a).
- **T3**: the split అడ విలో gives the phones of అడవిలో; no pause between the parts.
- **T4**: 2026 → రెండు వేల ఇరవై ఆరు; 12వ → పన్నెండవ; 10:45 → పది నలభై ఐదు.

Observations. All five sentences pass the frontend with every number expanded to Telugu words. Whisper large-v3's transcripts show what is recognised, with its weak Telugu in mind (copy-synthesis ceiling 33.1 %, Section 7): **T1**: the English loanword is recognised and written in Latin script (*Neuro-Pharmacological*), which is most of the CER; the rest is శ → స in పరిశోధనలో and విశ్లేషించారు and ఔ → ఉ in ఔషధాల. **T2a/T2b**: పట్టు is transcribed correctly in both senses (written and pronounced alike; only the prosody differs, Section 6.2); the errors are word-boundary merges (తాడు పయతని for తాడుపై అతని, ఆమేపట్టు) and vowel length (ఆమె → ఆమే, చీర → చిర). **T3**: the split అడ విలో is transcribed by Whisper as the single word అడవిలో (no pause between the parts, Section 6.2); the only error is again శ → స. **T4**: Whisper writes the spoken numbers back as digits (2026, 10.45), which the CER counts as errors, drops the opening ఈ సమావేశం, merges నవంబర్ పన్నెండవ తేదీ across word boundaries (the syllables are present) and hears ప్రారంభ without aspiration; the final ఉంచండి is exact.

Across the 22 distinct Telugu sentences scored (Sections 5.5, 7.3, 7.4; XT01 repeats T1) the same confusions recur: శ appears in the transcripts for only 2 of its 16 occurrences (mostly written స), ణ for 1 of 6 (written న), and 16 of 26 aspirated consonants are kept (కథ → కద, పాఠకులు → పాదకులు); some losses are words the ASR dropped entirely. The frontend keeps all these phones distinct, so the merger is in the acoustic model, in Whisper, or both; with this ASR the two cannot be separated *[interpretation]*.

6. Variance-adaptor plots, per test sentence

Each test sentence has three plots in `Results/<Lang>/variance_adaptor_plots/`:

- **Duration**: predicted frames per phone (bars), with milliseconds on the right axis.
- **Pitch**: predicted F0 per phone in Hz, drawn as a step over the phone's time span, together with the F0 that WORLD [7] extracts from the generated audio (grey dots; unvoiced frames are gaps). The overlay shows whether the decoder and vocoder realise the prediction.
- **Energy**: predicted energy per phone (L2 norm of the STFT frame, arbitrary units).

There is no ground truth for the test sentences, so this section describes what the model predicts; it does not evaluate it. For English, where inference-time options are used, each plot shows both the model's own prediction (hollow circles on the duration plot, dashed grey steps on pitch and energy) and the final values used for the audio (bars and solid black steps). The analysis describes the model's prediction first and then what the options changed. The numbers behind every plot are in `variance_raw/<ID>.json`; the per-word figures come from `tools/word_stats.py`.

Patterns common to all ten sentences (the model's own predictions)

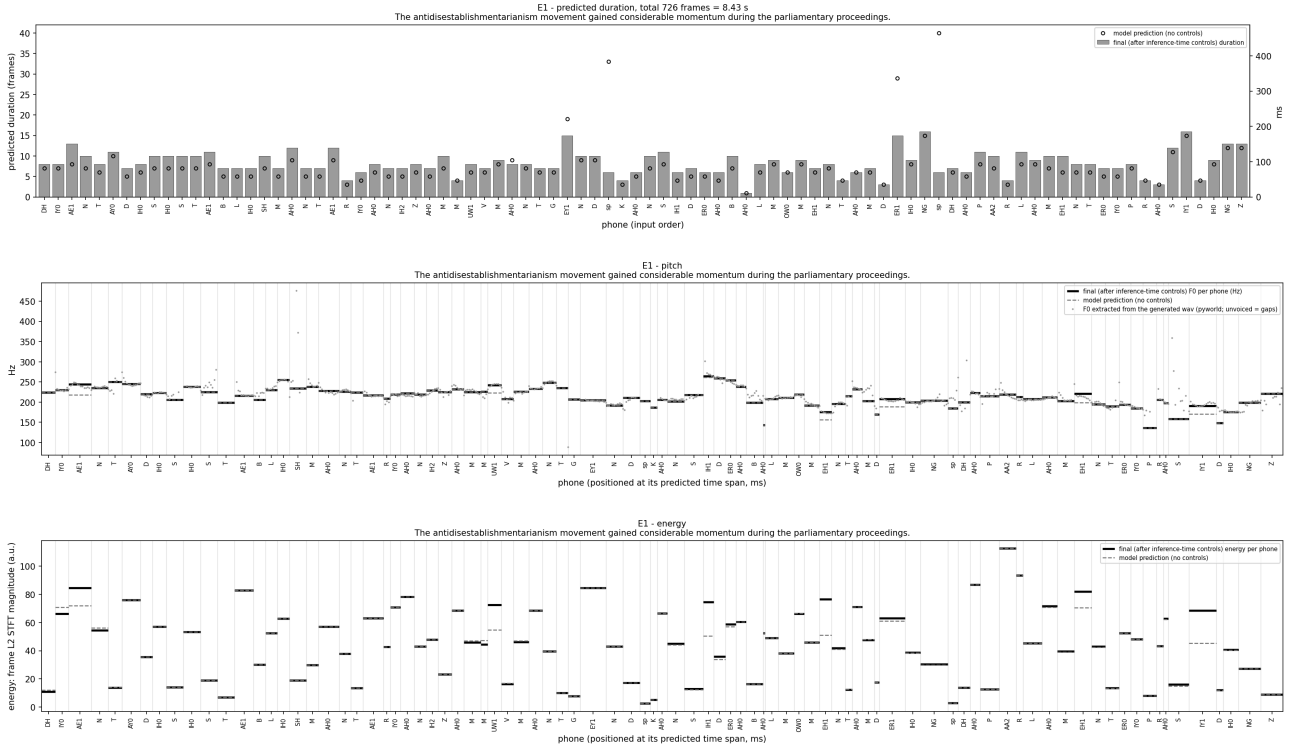
- **Declination**: the mean pitch of the last third of the phones is below that of the first third in every sentence (English 219–237 Hz → 195–209 Hz; Telugu 217–246 → 178–197 Hz). The exception to a falling end is E3, which the model ends on its highest pitch (269 Hz on *unexpectedly*), a question-like rise on a statement.
- **Phrase-final lengthening**: the last vowel is 1.1–2.5× the sentence's mean vowel duration in English (except E2a, 0.9×, whose last word ends in /s/) and 1.3–2.1× in Telugu.
- **Segmental energy pattern**: vowels and nasals high, stops, fricatives and pauses low (e.g. E4: /s/ and /z/ 4.5–14, vowels 30–123 a.u., median 64), as expected for an STFT-norm energy.
- **Speaking rate** 11.0–12.5 phones/s (English), 12.5–14.5 phones/s (Telugu).
- **Lexical stress (English)** is weakly marked: the noun and verb *conduct* get almost the same syllable durations (Section 6.1). This is what the data contain: in the Svarah training recordings the stressed vowel is the most prominent vowel of its word in 43.6 % of words, against 40.7 % expected by chance, and the model's predictions match the validation recordings (45.6 % vs 44.4 %). "Most prominent" = higher than every unstressed vowel of the word on at least two of pitch, duration and energy; words with a primary-stressed and an unstressed vowel; 1,074 training and 340 validation words, from the 428 training and 141 validation clips whose frontend phones match the forced alignment exactly (the other clips were skipped, so this is a subsample) (`fastspeech2/stress_stats.py`, Kaggle `fs2-15-stress-stats`).
- The F0 measured on the generated audio (WORLD, `generated_f0_hz`) follows the predicted phone-level F0: on voiced frames the median deviation is 27–51 cents per sentence (under half a semitone) and the frame-level correlation 0.50–0.97, so the decoder and vocoder realise the predicted pitch; in the plots the larger deviations sit at phone boundaries and on short phones, as expected from step-wise phone-level targets.

6.1 English

E1: The antidisestablishmentarianism movement gained considerable momentum during the

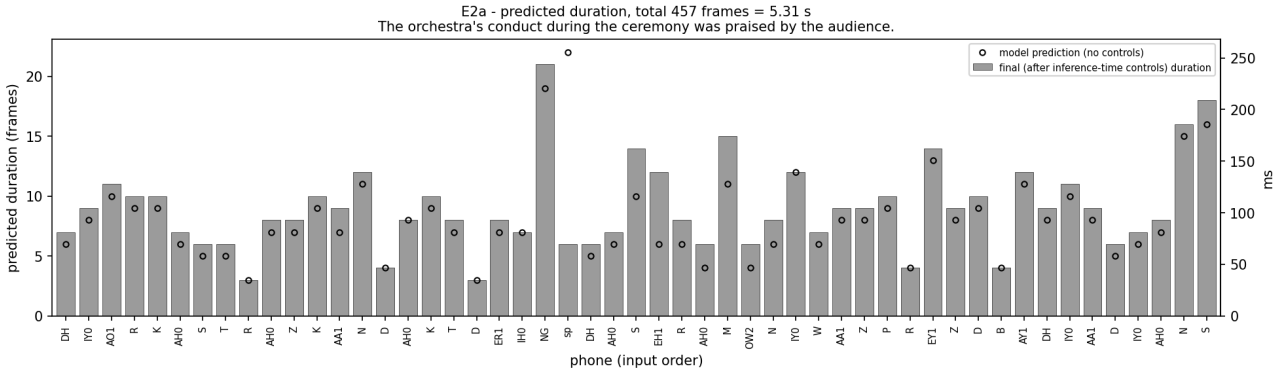
parliamentary proceedings.

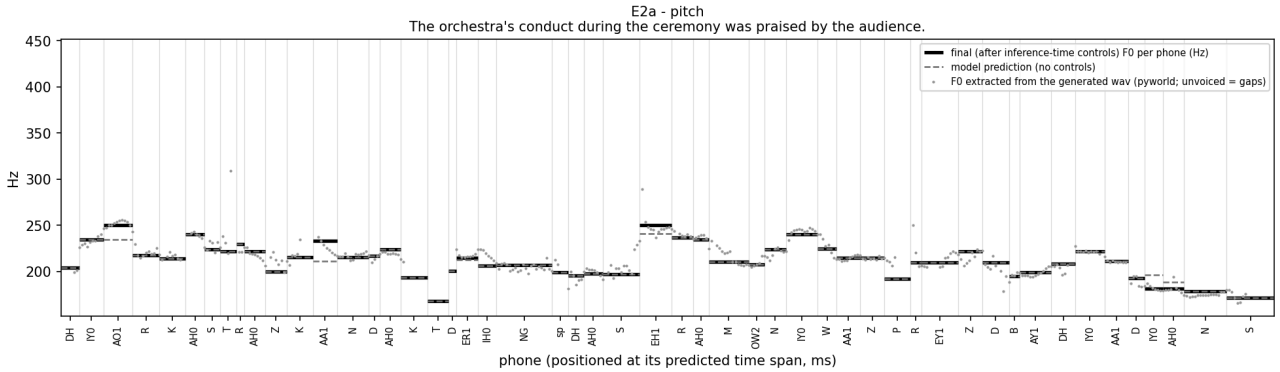
- **Duration (model).** *antidisestablishmentarianism* takes 2,020 ms (26 % of the model's 7.74 s) with uniformly short phones; *parliamentary* 801 ms, *proceedings* 882 ms. The final vowel is only 1.1× the mean vowel: little phrase-final lengthening. At the two micro-pause positions the model predicts 383 and 464 ms, as long as phrase-boundary pauses; the option sets them to 70 ms.
- **Pitch (model).** Several similar peaks (*considerable* 261, *antidisestablishmentarianism* 256, *movement* 248 Hz), lowest on *during* (204 Hz), first third 225 → last third 195 Hz: several peaks of similar height and an overall fall.
- **Energy (model).** Vowels median 63, nasals 43, other consonants 15 a.u., pauses ≈ 3; the final vowel of *proceedings* is 41.
- **Options.** Long words × 1.25 and everything × 1.10: the long word 2,020 → 2,798 ms, *parliamentary* 801 → 1,103 ms; the stress rule raised the stressed vowel of seven words (pitch up to × 1.12, energy up to × 1.5). Whisper recognises the long word exactly.

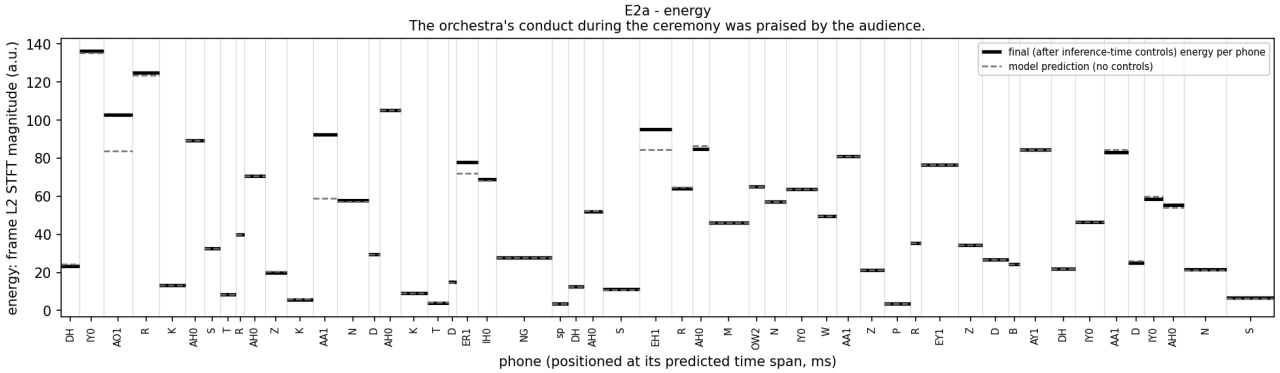


E2a: The orchestra's conduct during the ceremony was praised by the audience.

- **Duration (model).** *conduct* (noun, `K AA1 N D AH0 K T`): syllables 313 and 325 ms (ratio 0.96), the longest word of the sentence (639 ms). *audience* is long (662 ms) but its final vowel is short (0.9× the mean) because the word ends in /s/.
- **Pitch (model).** The stressed first vowel is *lower* than the unstressed second (211 vs 224 Hz): the model does not mark the noun's stress, in line with the data (Section 6, patterns). Main accents on *orchestra's* and *ceremony* (241 Hz); first third 220 → last third 201 Hz.
- **Energy (model).** Vowels median 74, nasals 46, other consonants 22 a.u.; highest on the /iː/ of the article before *orchestra's* (135).
- **Options.** The stress rule makes the first vowel of *conduct* the higher (233 vs 224 Hz) and the louder (× 1.5), with syllables 360 / 348 ms; the terminal fall lowers the last two phones to 182 Hz; the micro-pause before *the ceremony* is 70 ms (model: 255 ms).

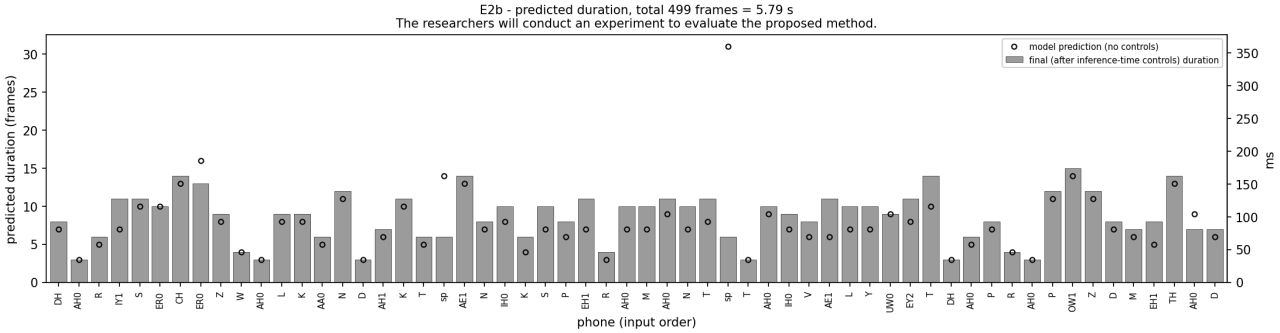


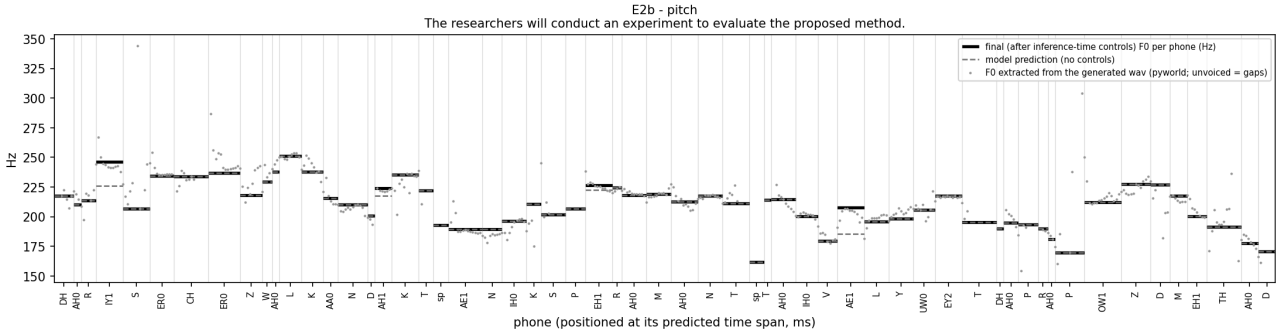


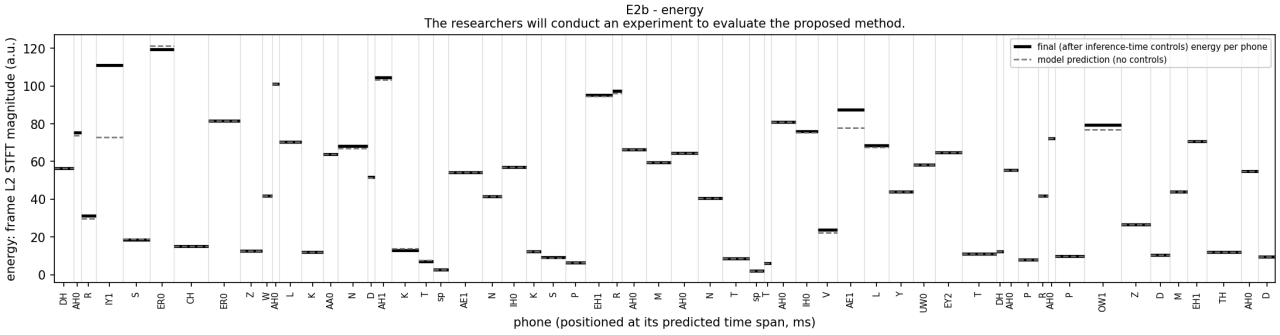


E2b: The researchers will conduct an experiment to evaluate the proposed method.

- **Duration (model).** *conduct* (verb, `K AA0 N D AH1 K T`; the input differs from the noun only in the stress digits): syllables 279 / 279 ms (ratio 1.00, against 0.96 for the noun) and 557 ms in all (noun: 639 ms). In the model's prediction the noun/verb difference is mainly the word's length, not which syllable is prominent. Final lengthening 1.2× on *method*.
- **Pitch (model).** Vowels of *conduct* 215 / 218 Hz (almost level); peaks on *will* (251 Hz) and *researchers* (237 Hz); lowest on *an* (189 Hz); first third 224 → last third 198 Hz.
- **Energy (model).** Vowels median 72, nasals 44, other consonants 12 a.u.; the strongest phone is the /ɚ/ of *researchers* (121).
- **Options.** The stress rule puts the higher pitch on the stressed second vowel (224 vs 215 Hz). With E2a (233 vs 224 Hz on the first vowel) the final audio therefore carries the homograph contrast in pitch, which the model alone hardly does. Whisper: 10.9 % CER with the post-filter only (configuration R0, earlier frontend: "an experimental evaluation of"), exact with the final options.

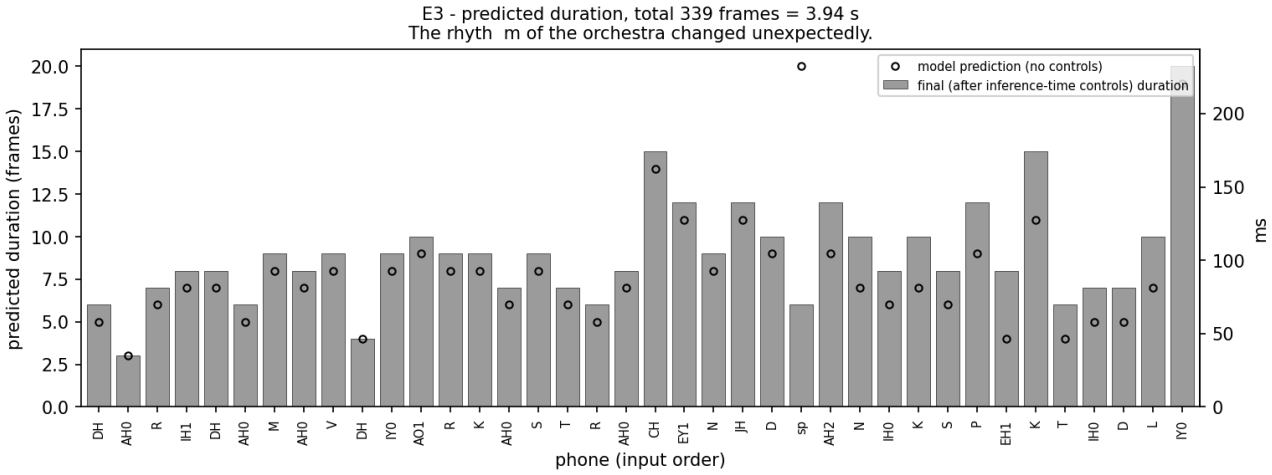


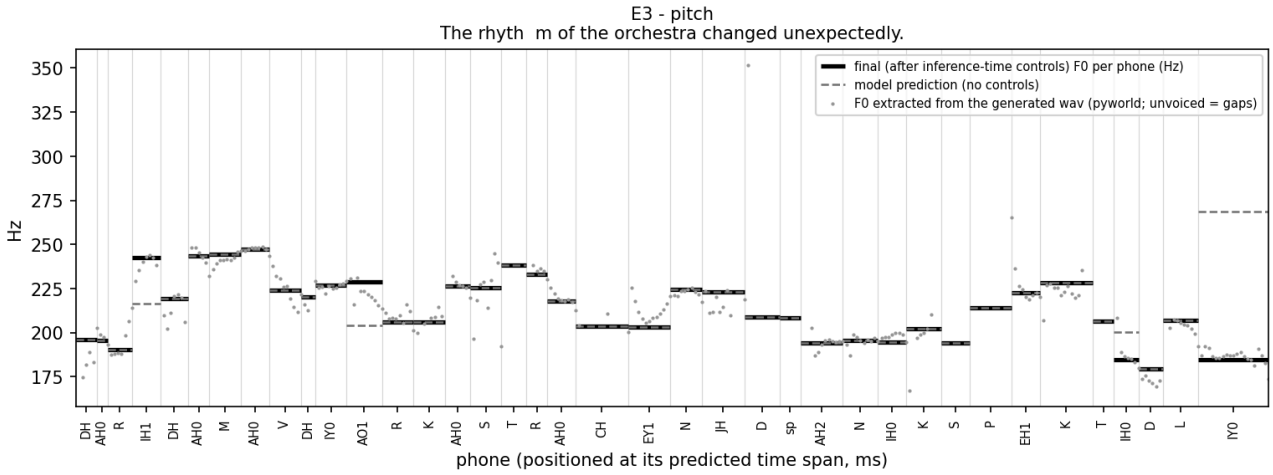


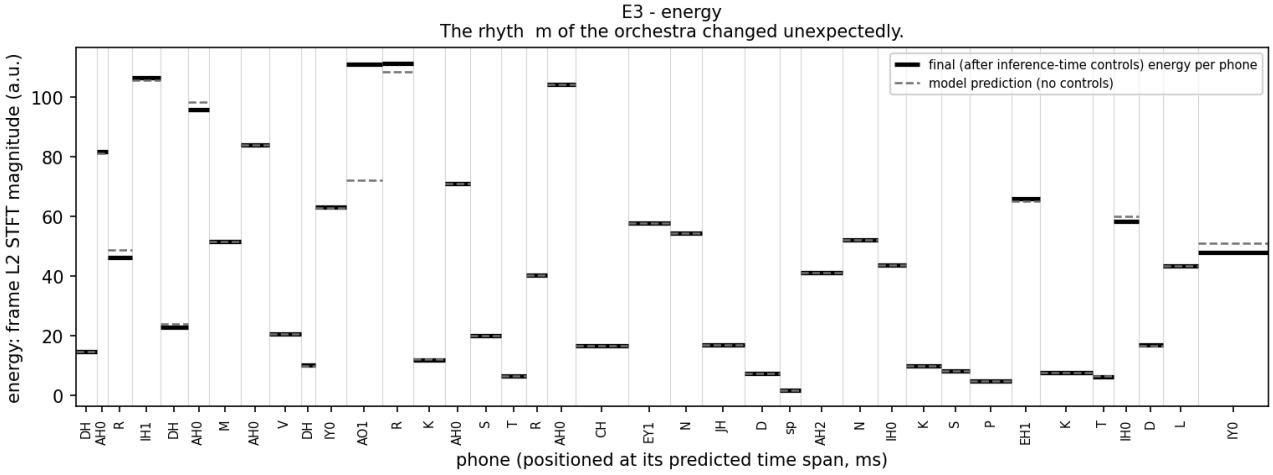


E3: The rhyth m of the orchestra changed unexpectedly.

- **Duration (model).** The split word is pronounced as *rhythm* (`R IH1 DH AH0 M`, 383 ms), without a pause inside it: the double space has no effect after joining. *unexpectedly* is very long (1,149 ms) with a final vowel of 221 ms (2.5× the mean), the strongest final lengthening of the five sentences.
- **Pitch (model).** The model ends this statement on its highest pitch: the last vowel of *unexpectedly* is 269 Hz, a question-like rise, while *rhythm* and *of* carry 244–248 Hz.
- **Energy (model).** Vowels median 68, nasals 52, other consonants 16 a.u.; the /r/ of *orchestra* is the energy maximum (108), as an approximant with vowel-like energy.
- **Options.** The terminal-fall rule lowers the last two vowels from 201 and 269 Hz to 185 Hz, turning the rise into a statement fall; *unexpectedly* 1,149 → 1,544 ms (long word); the stress rule adds energy to the first vowel of *orchestra* (× 1.5).



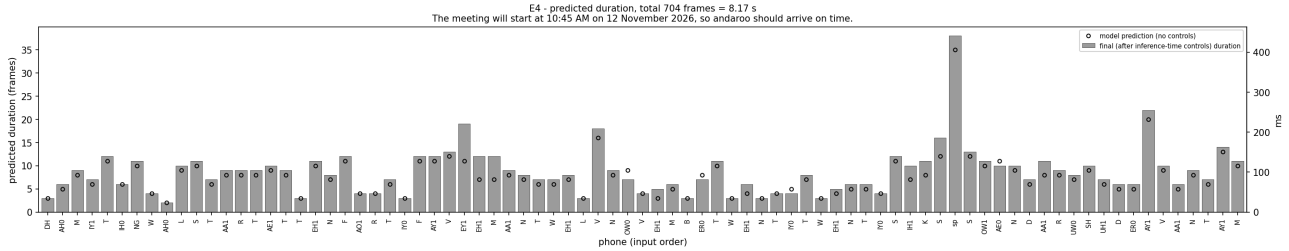


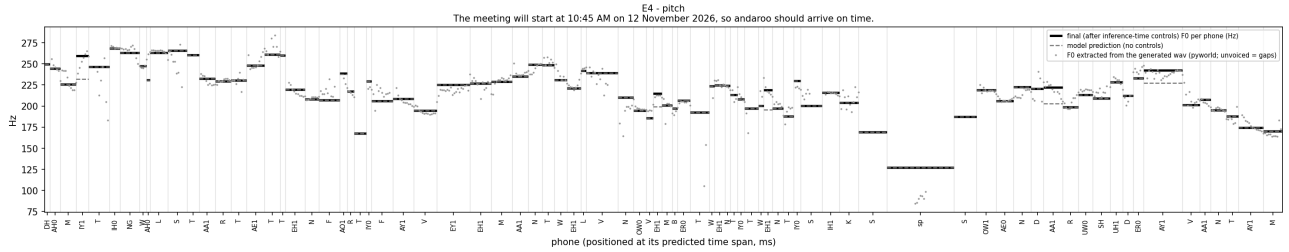


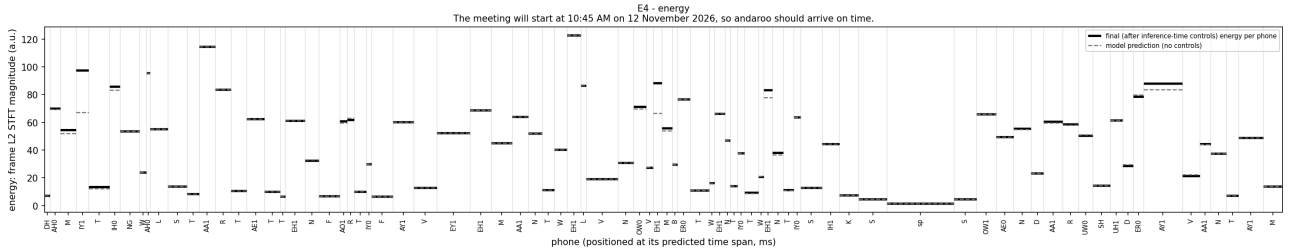
E4: The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

- **Duration (model).** Numbers read as words with growing durations (*ten* 232, *forty* 325, *five* 395 ms; *twelve* 441, *November* 464, *six* 441 ms): each number group ends on its longest word; *twenty twenty* 337 + 325 ms; *AM* 290 ms. The comma gives a 406 ms pause. After it the function words are very short (*should* 232 ms with a 70 ms vowel, *on* 151 ms); final lengthening 1.8× on *time*.
- **Pitch (model).** Highest on *meeting* (268 Hz), lowest on *time* (188 Hz); no pitch reset after the comma (*six* 216 → *so* 219 Hz), unlike Telugu T4; first third 237 → last third 205 Hz.

- **Energy (model).** Vowels median 64, nasals 46, other consonants 13 a.u.; /s/ of *six* and *so* ≈ 5 (fricatives), the comma pause 1.4.
- **Options.** *AM* as letter names × 1.6 (× 1.10): 290 → 499 ms; comma pause 441 ms; *twenty* in its citation form with /t/; the stress rule lengthens the stressed vowels of *November* (× 1.4), the first *twenty* (× 1.31) and *andaroo* (× 1.24). The last clause remains the weak point for the ASR (Section 5.4).



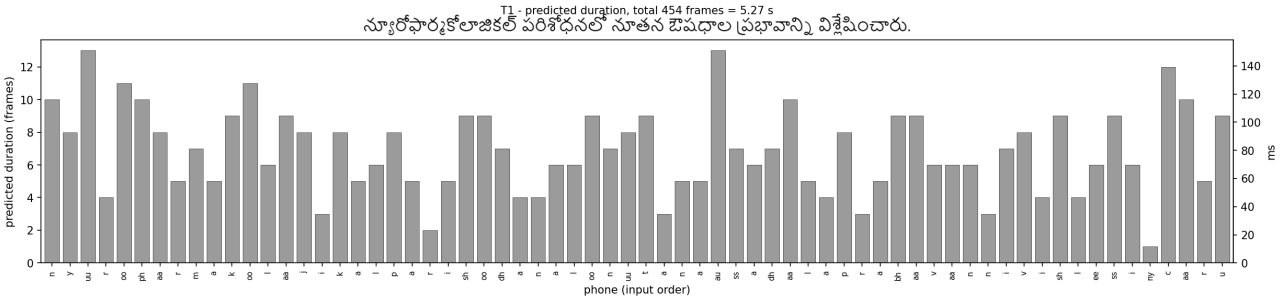


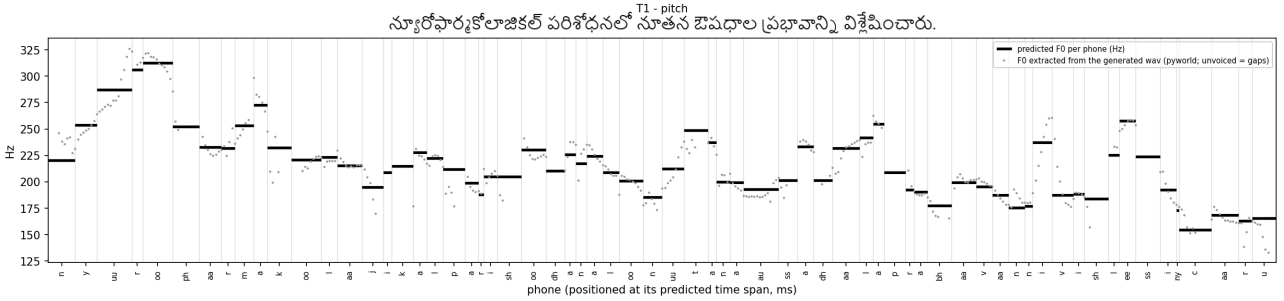


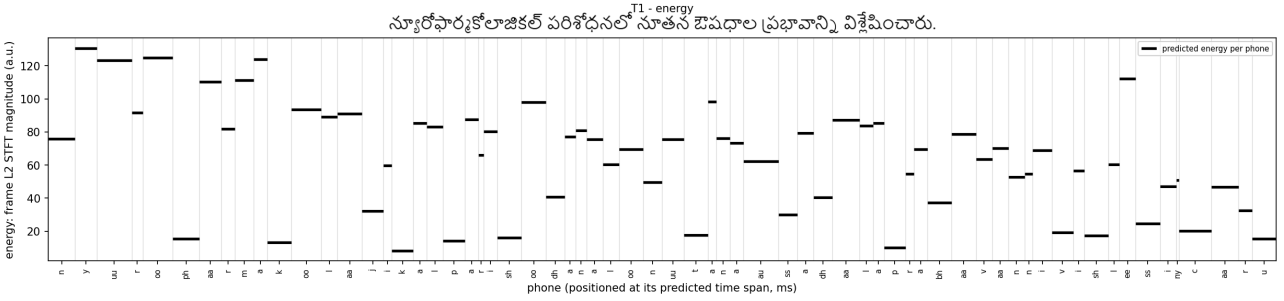
6.2 Telugu

T1: న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

- న్యూరోఫార్మకోలాజికల్ lasts 1,695 ms (32 % of 5.27 s) and carries the sentence's highest pitch (313 Hz).
- Pitch drops to 230 Hz on పరిశోధనలో and stays in a narrow 237–258 Hz band for the rest: the F0 maximum is early and the sentence-final verb is low.
- The final verb విశ్లేషించారు is long (964 ms; cluster `sh l`, nasal `ny` before `c`) with 1.3× final lengthening; no internal pauses (no internal punctuation).
- **Energy.** Vowels median 79, nasals 65, other consonants 35; the final /u/ of విశ్లేషించారు falls to 15: the energy falls over the last syllables.

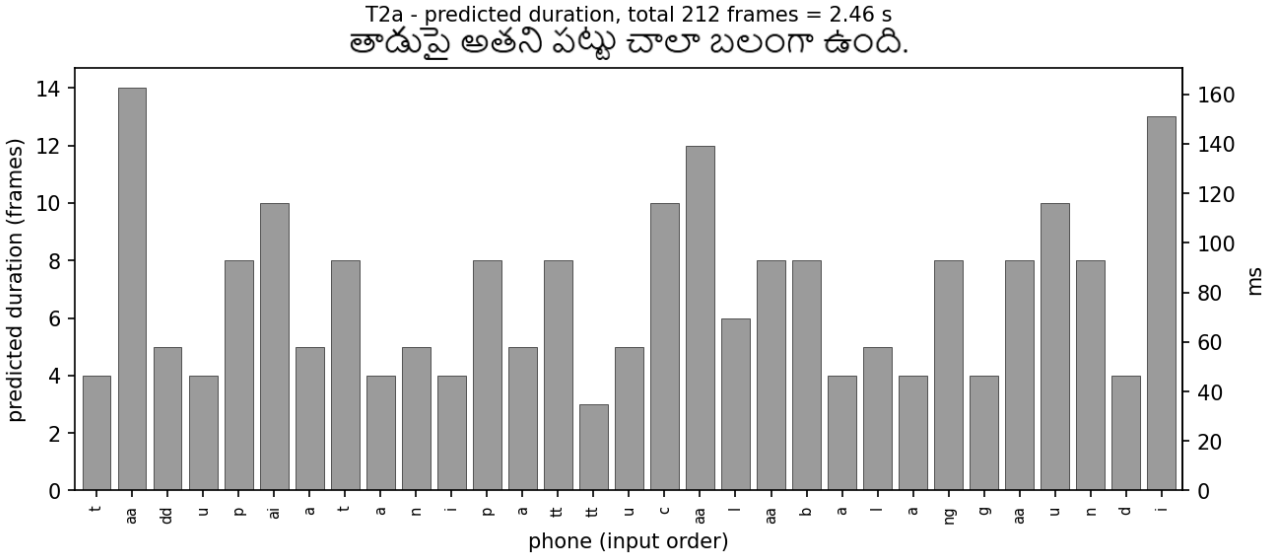


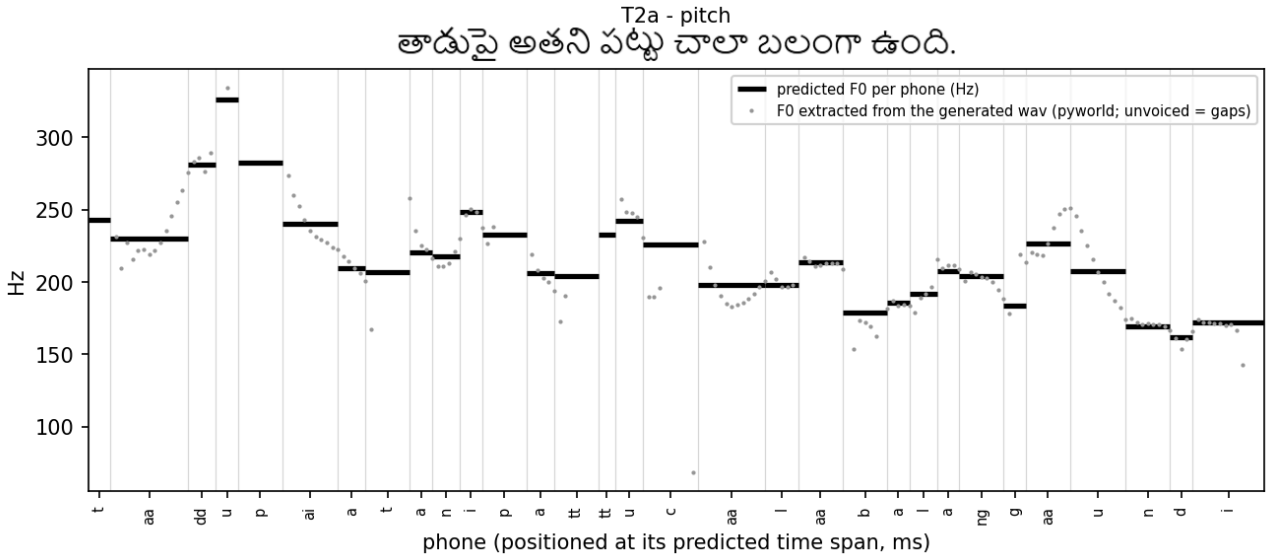


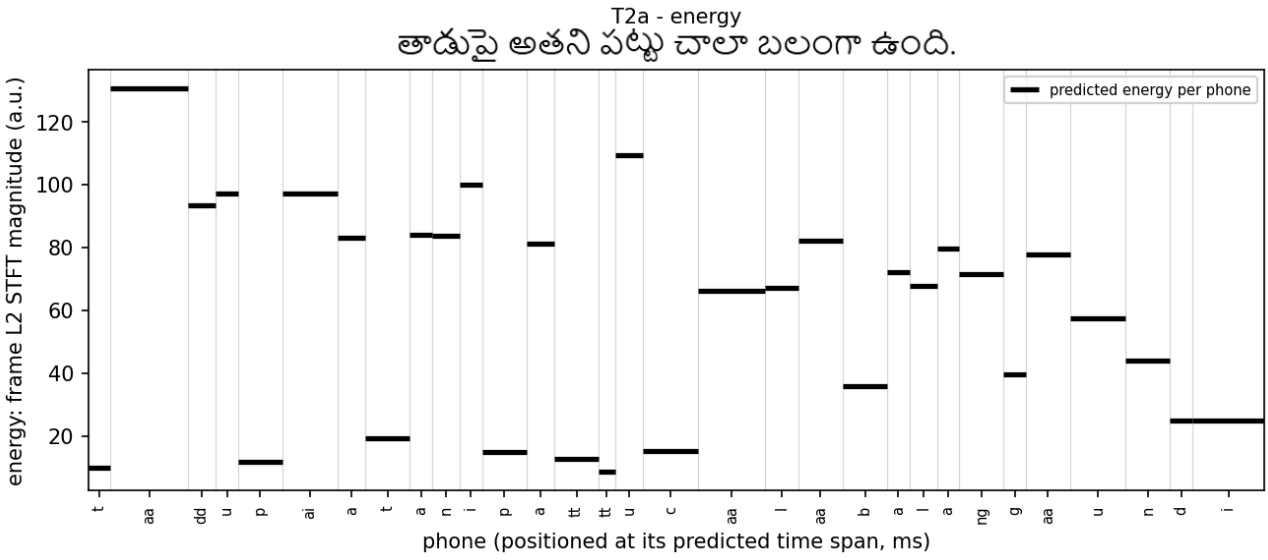


T2a: తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

- Peak pitch on the first word తాడుపై (326 Hz), then a steady fall to ఉంది (207 Hz).
- పట్టు (`p a tt tt u`, geminate retroflex) lasts 337 ms with 242 Hz, inside the falling line, not accented.
- Final vowel 151 ms, 1.8× the mean: strong phrase-final lengthening.
- **Energy.** Vowels median 82, nasals 72, other consonants 19; the geminate `tt tt` of పట్టు is the energy minimum (8.6), the long closure of the stop; the final /i/ drops to 25.

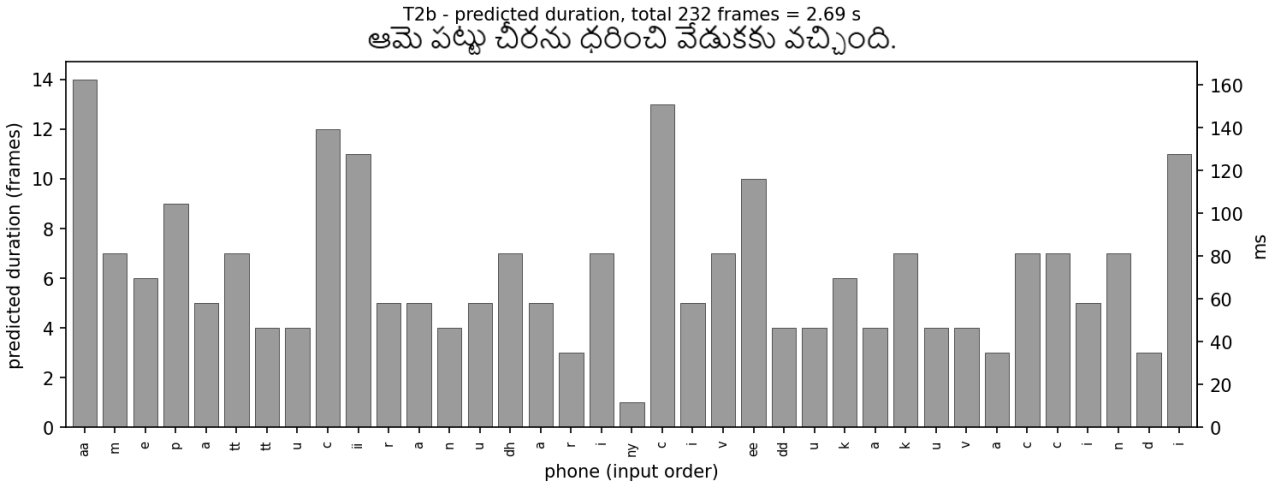

T2a - predicted duration, total 212 frames = 2.46 s
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
predicted duration (frames)
ms
phone (input order)
t aa dd u p ai a t a n i p a tt tt u c aa l aa b a l a ng g aa u n d i

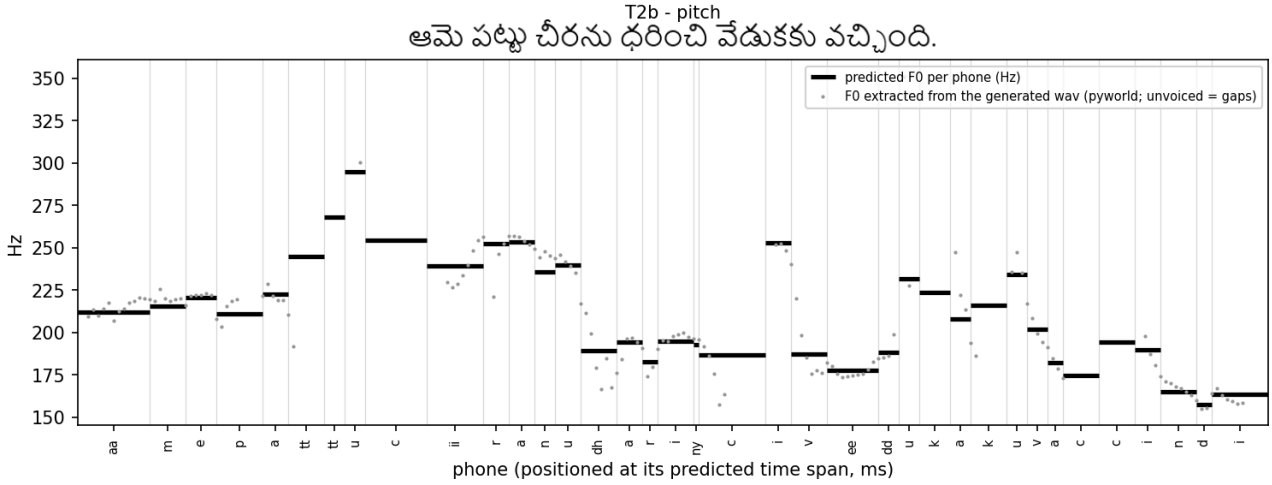

T2a - pitch
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
predicted F0 per phone (Hz)
F0 extracted from the generated wav (pyworld; unvoiced = gaps)
Hz
phone (positioned at its predicted time span, ms)

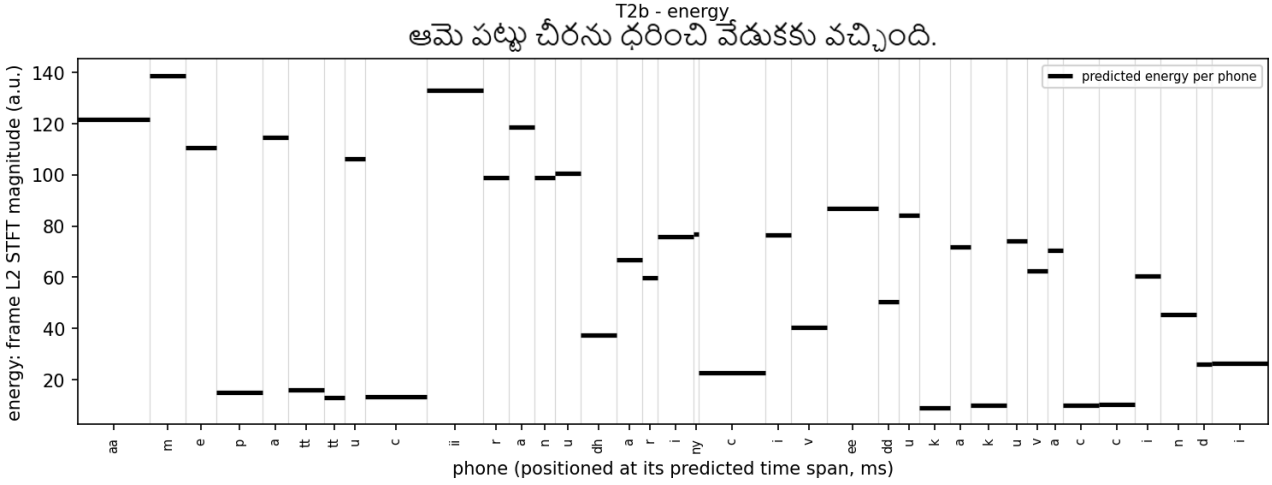

T2a - energy
తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
predicted energy per phone
energy: frame L2 STFT magnitude (a.u.)
phone (positioned at its predicted time span, ms)

T2b: ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

- పట్టు has the same phones and the same duration (337 ms) as in T2a, as it should (the two words are written and pronounced alike), but now carries the sentence's F0 maximum (295 Hz vs 242 Hz in T2a), directly before చీర and after a low start on ఆమె (220 Hz).
- The rest declines from 254 Hz (చీరను) to 202 Hz (వచ్చింది), with 1.7× final lengthening.
- The input phones of the word are identical in T2a and T2b; the model gives it the same duration and a different F0 depending on its position in the sentence (there is no recording of these sentences to compare with).
- **Energy.** Vowels median 84, nasals 88: the /m/ of ఆమె is the strongest phone (139), so the sentence starts loud; other consonants 19; the final /i/ drops to 26.



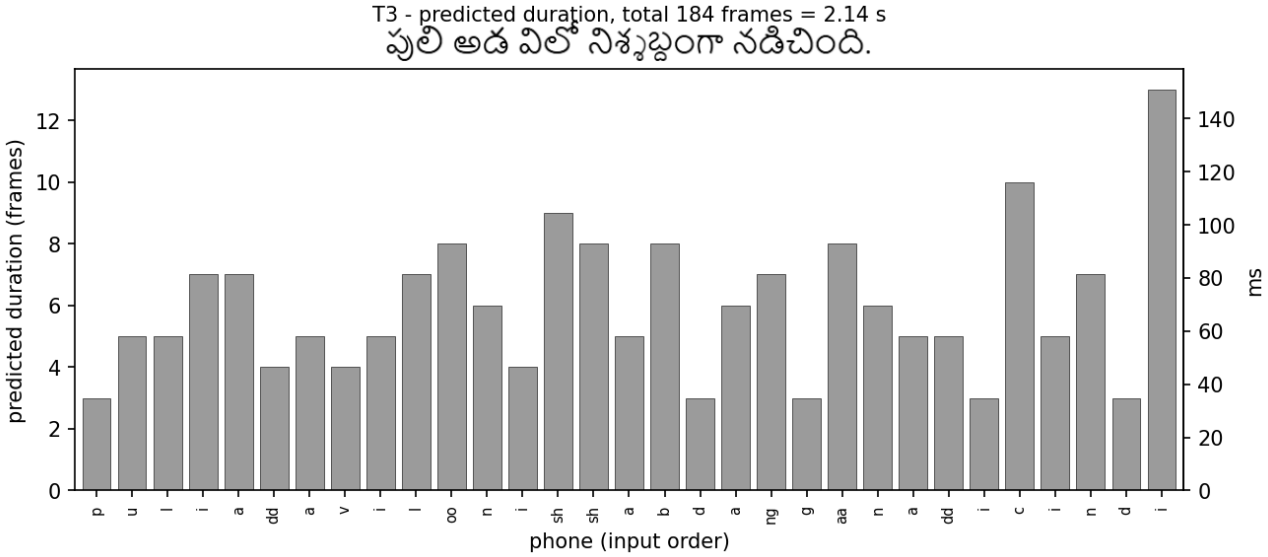


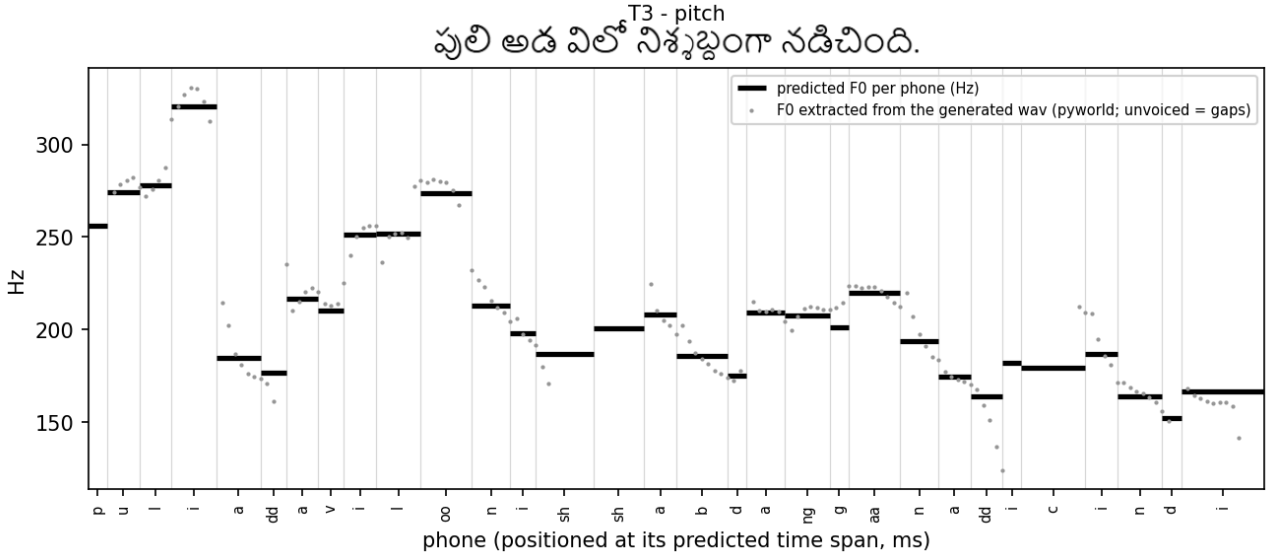


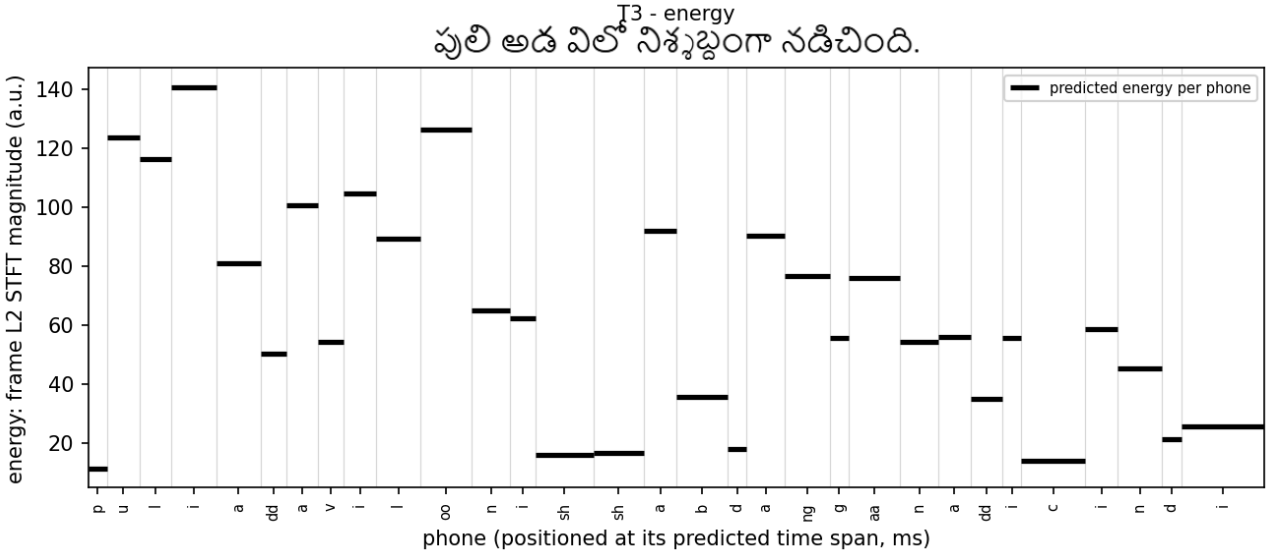
T3: పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

- Peak on పులి (321 Hz, the topic).
- The broken word అడ విలో gives exactly the phone sequence of అడవిలో (`a dd a v i l oo`; checked), because the Telugu input has no word-boundary symbol: the model receives the intended word. It is spoken without a pause (అడ 186 ms, విలో 279 ms, contiguous), with pitch rising from 217 to 274 Hz across the word.

- నిశ్శబ్దంగా is long (778 ms; geminate `sh sh` and the cluster `b d`); నడిచింది ends the sentence at the pitch minimum (194 Hz) with 2.1× final lengthening.
- **Energy.** Vowels median 86, nasals 60, other consonants 35; strongest on an /i/ (141); the final /i/ of నడిచింది drops to 26.



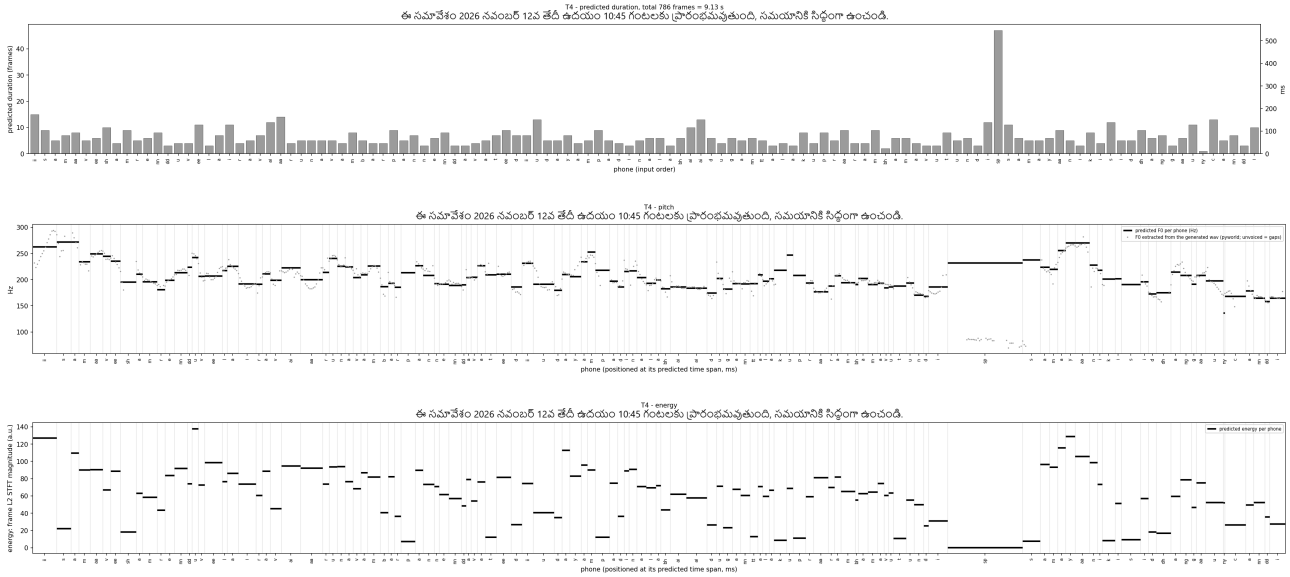




T4: ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది,

సమయానికి సిద్ధంగా ఉంచండి.

- Longest sentence (9.13 s, 120 phones). Numbers are expanded to Telugu words: రెండు వేల ఇరవై ఆరు (302/290/453/267 ms), పన్నెండవ (615 ms), పది నలభై ఐదు (244/418/267 ms), read without hesitation pauses.
- The comma gives a 546 ms pause followed by a clear **pitch reset**: from ≈ 208 Hz on ప్రారంభమవుతుంది (the longest word, 1,126 ms) up to 270 Hz on సమయానికి: the F0 of the second clause starts high again (a reset).
- Highest pitch on సమావేశం (272 Hz), lowest on the final ఉంచండి (198 Hz), with 1.5× final lengthening.
- **Energy.** Vowels median 75, nasals 74, other consonants 37; the comma pause is near silence (0.3); the final /i/ drops to 28. In all five Telugu sentences the last vowel has 15–28 against a vowel median of 75–86, a stronger end-of-sentence decay than in English (41–55 against 63–74).



7. Objective evaluation on held-out data

Both models were evaluated on **every validation utterance** (never trained on), each synthesised with its own speaker and without any inference-time option (`fastspeech2/evaluate_val.py`, Kaggle `fs2-04-eval`, `fs2-06-eval-fix`). Silence phones are excluded from duration/pitch/energy metrics.

Metric	English (742 utt.)	Telugu (446 utt.)
Duration MAE (frames, 11.6 ms) / Pearson r of log(1+d)	2.81 / 0.58	1.06 / 0.85
Total utterance length, relative error	11.7 %	4.5 %
Phone-level pitch RMSE / Pearson r	34.6 Hz / 0.35	29.2 Hz / 0.76
Phone-level energy RMSE / Pearson r	26.8 / 0.78	17.4 / 0.90
Mel-cepstral distortion*, ground-truth duration/pitch/energy (DTW)	36.6	26.1
Mel-cepstral distortion*, predicted duration/pitch/energy (DTW)	39.0	30.9
Whisper large-v3 CER / WER, our synthesis (150 utt.)	15.7 % / 21.5 %	35.8 % / 80.3 %

Same, copy-synthesis ceiling (ground-truth mel → same HiFi-GAN)	11.6 % / 13.8 %	33.1 % / 78.6 %

* Computed from the model's own 80-band log-mel (orthonormal DCT, c1–c24), averaged over DTW-aligned frames. This is not the WORLD/SPTK mel-cepstrum MCD [15] reported in papers; it is comparable only within this table. CER/WER are corpus-level (total edits / total reference length) after lower-casing and removing punctuation; CER ignores spaces. Whisper large-v3 is weak on Telugu: its CER on copy-synthesis of real Telugu validation recordings is 33.1 %, against 11.6 % for English, so Telugu CERs bound pronunciation errors from above rather than measure them. A Telugu ASR model (`vasista22/whisper-telugu-large-v2` [17], not trained on IndicTTS; pinned in `evaluate_val.ASR_REVISIONS`) was prepared for this evaluation (Kaggle `fs2-20-asr-te-cpu`), but it runs on CPU (the weekly GPU quota was used up) and had not finished when this report was built.

Reading the table. (1) The variance predictors add a small part of the distortion: replacing ground-truth duration, pitch and energy by predictions raises the distortion by 2.4 (en) and 4.8 (te); most of the remaining error is in the decoder itself. (2) Telugu prosody is predicted with much higher correlation than English (pitch r 0.76 vs 0.35; duration r 0.85 vs 0.58). A plausible reason is that two speakers reading prepared text vary less than Svarah's 110 speaker profiles; data, speakers and training length (3.1× more steps for Telugu) are confounded, so this is not established *[interpretation]*. (3) For English the intelligibility gap to the vocoder ceiling is 4.1 points of CER: the acoustic model loses some intelligibility, but most of the error (11.6 of 15.7 points) is already present in copy-synthesis of the real recordings. (4) For Telugu, Whisper large-v3 loses nearly as much on copy-synthesis of real recordings (33.1 %) as on our synthesis (35.8 %): as far as this ASR can tell, the acoustic model costs 2.7 points of CER.

7.1 Speed

Real-time factor (compute s / audio s), batch 1, warm	English	Telugu
GPU (Tesla T4): acoustic model / HiFi-GAN / total	0.0079 / 0.0193 / 0.027	0.0070 / 0.0186 / 0.026
CPU (4 threads, fp32, 20 utterances): acoustic / HiFi-GAN / total	0.029 / 0.322 / 0.350	0.026 / 0.339 / 0.366

On a T4 one second of speech takes 26–27 ms (37–39× faster than real time); the paper reports RTF 1.95×10^{-2} on a V100 with Parallel WaveGAN *[paper]*. On a CPU the acoustic model is still 35–38× faster than real time; the vocoder dominates (92–93 % of the time), so a lighter vocoder is the lever for CPU deployment. Timing excludes the text frontend; CUDA is synchronised before each reading; one untimed warm-up call.

7.2 Prosody controls

Each control factor was applied to one sentence per language (E2b and T2a), and the length, median F0 and RMS level of the audio were measured.

Control (factor)	English: length / median F0 / RMS	Telugu: length / median F0 / RMS
none	4.55 s / 206 Hz / 0.146	2.46 s / 209 Hz / 0.148
duration 0.8 / 1.25	3.37 s / 5.42 s	1.88 s / 3.02 s
pitch 0.8 / 1.2 (after M12)	169 Hz / 245 Hz	165 Hz / 252 Hz
pitch 0.8 / 1.2 (reference code)	140 / 113 Hz: inverted [c]	204 / 209 Hz: almost no effect [d]
energy 0.8 / 1.2 (after M12)	RMS 0.126 / 0.164	RMS 0.120 / 0.159

The duration control scales length close to the factor (0.74× and 1.19× for English; 0.76× and 1.23× for Telugu; rounding of per-phone frames explains the difference). Pitch control after M12 is near-proportional (×0.79/×1.20 for Telugu).

[c] Earlier English test voice, 126 Hz without control (`Evaluation/English/metrics_validation.json`). [d] From the console log of Kaggle job `fs2-04-eval`; not stored as a file.

7.3 Stress sentences

Sixteen further sentences, written for this project (not part of the assignment; `Evaluation/stress_sentences.json`), probe cases the test set does not: questions, homographs, names, abbreviations, numbers, conjuncts, code-mixing, very long and very short sentences. English uses the final configuration; Telugu the model as submitted.

English (final configuration; CER after Whisper's English normaliser, against the original text): corpus 1.7 %

ID	Case	Text	Whisper large-v3 transcript	CER norm.
SE01	time and numbers	The train leaves at 7:30 PM from platform 3.	The train leaves at 7.30 pm from Platform 3.	0.0 %
SE02	question intonation	Did you remember to bring the documents for tomorrow's interview?	Did you remember to bring the documents for tomorrow's interview?	0.0 %
SE03	homograph (wind: noun / verb)	The wind was strong, so we had to wind the rope tightly.	The wind was strong, so we had to wind the rope tightly.	0.0 %
SE04	Indian names and places	Priya and Venkatesh travelled from Hyderabad to Chennai last weekend.	Priya and Venkatesh travelled from Hyderabad to Chennai last weekend.	0.0 %
SE05	rare technical words	The cryptographic protocol relies on elliptic curve signatures.	The cryptographic protocol relies on elliptical signatures.	9.1 %
SE06	abbreviations and acronyms	Dr. Rao presented the results at IIIT Hyderabad.	Dr. Lau presented the results at IIT Hyderabad.	7.0 %
SE07	long sentence	Although the monsoon arrived later than expected this year, the farmers in the eastern districts managed to sow their crops on time, thanks to the new irrigation canals that were completed in early spring.	Although the monsoon arrived later than expected this year, the farmers in the eastern districts managed to sow their crops on time, thanks to the new irrigation canals that were completed in early spring.	0.0 %
SE08	exclamation, very short	What a wonderful surprise!	What a wonderful surprise!	0.0 %

SE05: "elliptic curve" heard as "elliptical"; SE06: the spelled acronym IIIT heard as "IIT" and *Rao* as "Lau".

Telugu (CER against the normalised text): corpus 18.9 %

ID	Case	Text	Whisper large-v3 transcript	CER, script-mapped

ST01	question intonation	మీరు రేపు ఉదయం ఆఫీసుకు వస్తారా? Will you come to the office tomorrow morning?	మీరు రేపు ఉదైయమాఫి సుకువస్తారా.	15.4 %
ST02	numbers	ఈ పుస్తకం ధర 250 రూపాయలు. The price of this book is 250 rupees.	ఈ పుస్తకం ధర రేండు వందల యాభి రూపాయలు.	6.7 %
ST03	time	రైలు సాయంత్రం 6:30 కి బయలుదేరుతుంది. The train departs at 6:30 in the evening.	రేలు సాయంత్రమారు ముప్పైకి బయలు దేరుతుంది.	8.3 %
ST04	consonant clusters (conjuncts)	స్త్రీ శక్తి రాష్ట్ర అభివృద్ధికి ముఖ్యం. Women's strength is important for the state's development.	3. సక్తి రాష్ట్ర ఆభివరుద్ధి కి ముఖ్యమ్	37.1 %
ST05	English loanwords in Telugu script (with ZWNJ)	నా లాప్‌టాప్ బ్యాటరీ త్వరగా అయిపోతోంది. My laptop battery is draining quickly.	4 లాప్టాప్ బ్యాటరి త్వరగా ఆయి పోతోంది.	12.1 %
ST06	Latin-script word inside Telugu (code-mixing)	నేను నిన్న కొత్త iPhone కొన్నాను. I bought a new iPhone yesterday.	నేను నిన్న కొత్త ఆఈ ఫోన్ కొన్నాను.	21.4 %
ST07	long sentence	హైదరాబాద్ నగరంలో ప్రతి సంవత్సరం జరిగే పుస్తక ప్రదర్శనకు వేలాది మంది పాఠకులు, రచయితలు మరియు విద్యార్థులు ఉత్సాహంగా హాజరవుతారు. Thousands of readers, writers and students enthusiastically attend the book fair held every year in Hyderabad.	AIDARABADNAGARAN లో ప్రతి సంవచ్రం జరిగే పుస్తక ప్రదర్శనకు వేలాది మంది పాదకులు, రచైతలు మరియు విద్యార్థులు ఉత్సాహంగా హాజరవుతారు.	20.2 %
ST08	exclamation, very short	ఎంత అందమైన దృశ్యం! What a beautiful sight!	ఎంతా అందమైన ద్రుస్యం.	33.3 %

ST06: the Latin-script *iPhone* is transcribed ఆఈ ఫోన్, i.e. spoken as intended through `text/telugu/latin.py`. Whisper writes digits where it hears none ("3." for స్త్రీ in ST04, "4" for నా in ST05) and drops question and exclamation marks (ST01, ST08), which the CER ignores.

7.4 Extended sentence set

Ten more sentences per language, supplied by the author (not part of the assignment; `Evaluation/Extended/extended_sentences.json`), extend each test sentence's difficulty (long technical words, homographs, numbers and dates, clusters, questions and exclamations, long sentences). XE01 and XT01 are the test sentences E1 and T1 verbatim, XT04 is the first clause of T4, and XE02/XT02 combine the two E2/T2 sentences; several were used while developing the frontend rules (Section 1.2), so they are not a blind test.

English: corpus CER 2.1 %

ID	Case	Text	Whisper large-v3 transcript	CER norm.

XE01	Long, uncommon words	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	The anti-disestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	0.0 %
XE02	Homographs, stress	The orchestra's conduct during the ceremony was praised, while the researchers will conduct another experiment tomorrow.	Bay Orchestra's conducting the ceremony was praised, while the researchers will conduct another experiment tomorrow.	5.8 %
XE03	Sibilants, articulation	She sells seashells beside the seashore, but the shells she sells are surely not the same.	She sells seashells beside the seashore, but the shells she sells are surely not the same.	0.0 %
XE04	Numbers, time, dates	At 10:45 AM on November 12, 2026, approximately 1,247 researchers will gather for the international conference.	At 10.45 a.m. on November 12, 2026, approximately 1,247 researchers will gather for the international conference.	0.0 %
XE05	Consonant clusters, pronunciation	The rhythmic patterns of the extraordinarily talented musician changed unexpectedly during the performance.	The rhythmic patterns of the extraordinarily talented musician changed unexpectedly during the performance.	0.0 %
XE06	Abbreviations, decimals	Dr. Williams asked whether the temperature would reach 37.5 degrees Celsius by 8:30 p.m.	Dr Williams asked whether the temperature would reach 37.5°C by 8.30pm.	17.6 %
XE07	Long sentence, prosody	Although artificial intelligence can process millions of calculations per second, human creativity remains remarkably difficult to replicate.	Although artificial intelligence can process millions of calculations per second, human creativity remains remarkably difficult to replicate.	0.0 %
XE08	Homographs, tense-dependent pronunciation	I read the book yesterday, and today I will read another chapter before going to sleep.	I read the book yesterday and today I will read another chapter before going to sleep.	0.0 %
XE09	Questions, emotional intonation	Wait! Are you seriously suggesting that we should rebuild the entire system from scratch?	Wait, are you seriously suggesting that we should rebuild the entire system from scratch?	0.0 %
XE10	Phonetic coverage, fluency	The quick brown fox jumps over the lazy dog while five extraordinary musicians quietly rehearse their final symphony.	The quick brown fox jumps over the lazy dog while five extraordinary musicians quietly rehearse their final symphony.	0.0 %

XE06's remaining CER is formatting: the ASR writes "37.5°C", which its normaliser does not map to "degrees Celsius"; the content is right. XE02: "The orchestra's conduct during" heard as "Bay Orchestra's conducting". Audio: `Evaluation/Extended/English/`.

Telugu: corpus CER 17.1 %

ID	Case	Text	Whisper large-v3 transcript	CER, script-mapped

XT01	Long Sanskrit-derived technical vocabulary	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	Neuro-Pharmacological పరిసోధనలో నూతన ఉషధాల ప్రభావాన్ని విస్లేశించారు.	38.7 %
XT02	Homographs: పట్టు	తాడుపై అతని పట్టు చాలా బలంగా ఉంది, కానీ ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	తాడు పయతని పట్టు చాలా బలంగా ఉంది, కాని ఆమే పట్టు చిరను ధరించి వేడుకకు ఉచ్చింది.	9.1 %
XT03	Consonant combinations, long sentence	పులి అడవిలో నిశ్శబ్దంగా నడిచింది, కానీ దాని గర్జనతో అడవి మొత్తం ఒక్కసారిగా ఉలిక్కిపడింది.	పులి అడవిలో నిస్చబ్దంగా నడిచింది. కాని దాని గర్జనతో అడవి మొత్తం మొక్క సారిగా ఉలిక్కి పడింది.	6.5 %
XT04	Dates, numbers, time	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది.	2026 నవం బర్పన్నెండవతేది ఉదైయం 10.45 గంటలకు ప్రారంభమోతుంది	50.0 %
XT05	క్ష, శ, త్ర combinations	క్షణక్షణానికి మారుతున్న పరిస్థితులను గమనిస్తూ శాస్త్రవేత్తలు పరిశోధనను కొనసాగించారు.	క్షణక్షనానికి మారుతున్న పరిస్తితులను గమనిస్తు సాస్త్ర వేత్తలు పరిసోధన అనుకొన సాగించారు. [Kannada script, mapped]	7.8 %
XT06	Compound words, pronunciation	హైదరాబాద్‌లోని విశ్వవిద్యాలయంలో కంప్యూటర్ సైన్స్ విద్యార్థులు కృత్రిమ మేధస్సుపై చర్చించారు.	AIDARABAD లోని విస్వవిద్యాలైయం లోకం కూప్యూటర్ సైంస్ విద్యార్ధులు కృత్రిమమే ధస్సుపై చర్చించారు.	19.5 %
XT07	Natural conversational Telugu	అమ్మ చెప్పిన కథలోని చిన్నారి ప్రతిరోజూ ఉదయాన్నే పక్షుల కిలకిలారావాలను వింటూ పెరిగింది.	అమ్మా చెప్పిన కదలోని చిన్నారి ప్రతి రోజు ఉదైయాన్నే పక్షులకి లక్కిలా రావాలను వింటు పెరిగింది.	9.2 %
XT08	Prosody, sentence rhythm	వర్షం పడుతున్నప్పటికీ, రేపటి ప్రయాణం కోసం మేము అవసరమైన వస్తువులన్నింటినీ సిద్ధం చేసుకున్నాము.	వర్షం పడుతున్నప్పటికి రేపటి ప్రయానం కోసం మేము అవసరమైన వస్తువుల అన్నింటి ని సిద్ధం చేసుకున్నాము.	4.9 %
XT09	Question intonation, emphasis	నిజంగా మనం సాధించాలనుకున్న లక్ష్యాన్ని చేరుకోవడానికి క్రమశిక్షణ, సహనం, పట్టుదల అవసరం కాదా?	నిజంగా మనం సాధించాలను కున్న లక్షాన్ని చేరుకోవడానికి క్రమ సిక్షన, సహనం, పట్టుదల అవసరం కాదా. [Kannada script, mapped]	5.1 %
XT10	Repeated syllables, long-form pronunciation	శ్రీకాకుళం నుంచి తిరుపతికి వెళ్లే రైలులో ప్రయాణిస్తున్న ప్రయాణికులకు రేపు ఉదయం ప్రత్యేక ప్రకటన చేయబడుతుంది.	స్రికా కుళ్ళమ్ముంచి తిరుపతికి వెల్లె రైలుల్లో ప్రయానిస్తున్న ప్రయానికులక్కు రేపు ఉదైయం ప్రత్తెక్క ప్రకటన చేయవడుత్తుంది. [Kannada script, mapped]	23.2 %

Whisper wrote XT05, XT09 and XT10 in Kannada script (76–107 Kannada characters each; raw CER 100–113 %); the table scores them after mapping the Kannada block back onto the Telugu block (same layout, offset 0x80; `tools/build_report.py`). XT09's question particle కాదా is recognised; its intonation falls (Section 8.2). XT04 and XT01 repeat T4's and T1's ASR effects (digits, Latin script). Whisper transcripts: `Evaluation/Telugu/metrics_extended_whisper.json` (Kaggle `fs2-14-extended`).

8. English vs Telugu; limitations, failures and improvements

8.1 Comparison

	English	Telugu
Data character	110 speaker profiles, ~2.7 min each, Indian-accented, recording conditions vary by speaker	2 speakers, ~2.5 h each, read speech
Text → phones	irregular spelling: lexicon + neural G2P; homographs, out-of-vocabulary words and normalisation are the main risks	near-phonemic script: deterministic rules cover every word; risks are normalisation and loanwords
Prosody prediction (validation)	lower: pitch r 0.35, duration r 0.58; lexical stress barely marked, as in the data	higher: pitch r 0.76, duration r 0.85
Over-fitting	mild (validation total lowest at 7k, +1.3 % by 17k)	minimal (total +0.2 % after 16k; duration term +3.7 %)
Output quality without inference-time help	over-smoothed decoder, recording noise reproduced, endings cut; needs the post-filter and clear-speech options (Sections 5.2–5.3)	not over-smoothed; used as the model produces it
Phrase structure	pause at the E4 comma with no pitch reset after it in the model's prediction	pause at the T4 comma followed by a clear pitch reset (208 → 270 Hz)
Main failures on the test set	E4's last clause (a made-up word followed by very short function words) is garbled for the ASR; short unstressed function words; lexical stress, question rises and statement falls come from rules, not from the model (Sections 5.3, 6); the muffled first test voice (Section 5.1)	శ heard as స, ఐ as స, lost aspiration (model or ASR, not separable here); word-boundary merges; T4's spoken numbers written back as digits by the ASR; no question intonation (XT09, Section 8.2)

8.2 Limitations

- **5 h of data** per language (the assignment's budget) against 24 h in the paper; the variance predictors show the largest train/validation gap (Section 4), and English prosody prediction on held-out data is weak (pitch r 0.35).
- **No listening test**: quality statements rely on ASR and acoustic measurements.
- **English decoder over-smoothing** (Section 5.2): the GV post-filter restores the spectral variance but only part of the frame-to-frame sharpness; the remaining smoothness is in the trained decoder; why it is larger for English than for Telugu was not tested.
- **The English output relies on inference-time options** (Section 5.3); the model alone (`Evaluation/English/model_only/`) is less clear. They are rules, not learned: they fire on punctuation and word length, not on meaning.
- **No question intonation in Telugu**: the Telugu model, used without rules, ends the yes/no question XT09 (…అవసరం కాదా?) with a falling, fading end (≈ 170 Hz on the last syllable against a sentence median of 206 Hz; Praat on `Evaluation/Extended/Telugu/XT09_te.wav`), i.e. with statement intonation; the "?" is not part of the model input.
- **Telugu G2P is orthographic**: it transcribes the script; it does not model schwa reduction, loanword pronunciations (ఫ as /f/), or sandhi. It was not reviewed by a native speaker.

- **Universal vocoder**: copy-synthesis costs 11.6 % CER on English validation speech; recording quality varies by Svarah speaker and bounds what each voice can sound like.
- **Telugu clips that MFA could not align were dropped** (28); their transcripts were not inspected.
- **Validation shares speakers with training** in both languages (split by recording / sentence, Section 2.1).
- **Svarah speaker identity is a metadata proxy** (110 profiles vs 117 speakers on the dataset card, Section 2.1).

8.3 Possible improvements (from the test sentences)

- **Voice and data selection by recording quality** (Section 5.1): select or weight training speakers by measured clarity, not only by amount of audio; the change of test voice (made in the same run as the letter-name durations for AM/PM) lowered the English test-sentence CER from 15.8 % to 2.7 % (Whisper-normalised, five sentences; commits 8f5ae5f → 1fcb7d6).
- **Word-boundary and sentence-type symbols in the input**: neither frontend marks word boundaries or the sentence type (as in the reference), so the model cannot separate "అడ విలో" from "అడవిలో", cannot learn the rise of a question (XT09) and ends E3 with a rise; adding such symbols needs retraining *[interpretation]*.
- **Early stopping / checkpoint selection** on validation loss (English: validation total lowest at 7k steps).
- **Against over-smoothing** (largest remaining English gap): a decoder loss that does not average (mixture output or an adversarial post-net), which needs retraining; or a longer vocoder fine-tune that starts from trained discriminators *[interpretation]*. Speaker adaptation needs more than 2.7 min of the target voice, or parameter-efficient adaptation (conditional layer norm, as in AdaSpeech [11]).
- **Learned instead of rule-based clarity**: training on clean, single-speaker English data of the same size would remove the need for denoising and edge handling; slower, clearer speech could be learned from data with that style.
- **Telugu**: native-speaker review of the G2P rules; question and exclamation marks as input symbols; frame-level pitch with CWT (as in the paper) for richer intonation.

8.4 Assumptions

- Telugu spelling is close enough to pronunciation for rule-based G2P. Supported, not proven: MFA aligned 98.9 % of the selected Telugu clips with the G2P dictionary; no native-speaker review.
- The universal HiFi-GAN suits these voices. Measured: copy-synthesis within ∼1 dB of the recordings per band up to 8 kHz; ASR ceilings 11.6 % (English) and 33.1 % (Telugu, weak ASR).
- MFA alignments are an adequate duration reference. Partly checked for Telugu (Section 2.2: 95 % of phones within one frame between two aligners).
- ASR character error is a usable proxy for intelligibility. Reasonable for English; weak for Telugu (Section 7).
- "About 5 h / 1 h" is counted on the audio of the used clips; trimmed hours are also reported (Section 2.1).
- The PDF text layer is the authoritative test text (Section 5).

8.5 What this report does not establish

- Naturalness or perceived quality: no listening test was run; statements marked as the author's impressions are not results.
- Whether the Telugu confusions (శ/స, ఐ/ఎ, aspiration) come from the acoustic model, the vocoder or Whisper.
- Why Telugu prosody is predicted better than English: language, speakers, data quality and the 3.1× more Telugu training steps are confounded.
- Generalisation to unseen speakers (validation shares speakers with training in both languages) or to other text domains.
- The cause of the E4 last-clause failure.
- Variation between training runs (one run per language, one seed) and the validation loss of the released steps themselves (last validation at 17k / 55k steps).
- That the English inference options would be chosen again on unseen text: they were compared on the test sentences plus 18 others, so the English test-sentence CER is optimistic.

9. Reproducibility

Software (as run on Kaggle): Python 3.13, PyTorch 2.11.0 (CUDA 12.8 on T4; CPU build for data jobs), librosa, pyworld, tgt, g2p_en, MFA 3.4.3 (conda-forge, via micromamba), transformers 5.16.1 (validation evaluation) and 5.20.2 (later jobs; evaluation only). Full package lists are in each job's `environment.txt`; `requirements.txt` lists the minimum versions. Data: Svarah and IndicTTS Telugu from Hugging Face at pinned revisions (`data/ingest.py`); ASR models at pinned revisions (`evaluate_val.ASR_REVISIONS`).

Configuration: `English/config.yaml`, `Telugu/config.yaml` (preprocess / model / train sections; step values as used), and the exact configs as run in `Results/<Lang>/train_config_used.yaml`.

Commands (data root `D`; Telugu shown, then the English lines that differ):

```
cd fastspeech2
python -m data.ingest --lang te --cache /tmp/hf --out D        # needs a Hugging Face token
python -m data.select_split --manifest_dir D/manifests/te --train_hours 5 --val_hours 1 --balance equal
# English: --manifest_dir D/manifests/en (default --balance proportional)
python data/mfa_prepare.py --lang te --ingest_dir D --corpus /tmp/corpus_te \
       --dict_out D/mfa_te/dictionary.txt
mfa train /tmp/corpus_te D/mfa_te/dictionary.txt D/mfa_te/model.zip \
       --output_directory D/preprocessed/te/TextGrid
# English: mfa align /tmp/corpus_en D/mfa_en/dictionary.txt english_us_arpa \
#          D/preprocessed/en/TextGrid --beam 100 --retry_beam 400
cd ..
python Telugu/preprocess.py --data_root D
python Telugu/train.py      --data_root D --max_minutes 480
python Telugu/inference.py  --data_root D --checkpoint Telugu/checkpoints/te_model.pth.tar \
       --speaker te_female --out_dir Results/Telugu
python fastspeech2/evaluate_val.py --config Telugu/config.yaml --data_root D \
       --checkpoint Telugu/checkpoints/te_model.pth.tar --speaker te_female --out_dir Evaluation/Telugu

# English (pretrained MFA aligner above; 210-minute budget = 17,949 steps)
python English/preprocess.py --data_root D
python English/train.py      --data_root D --max_minutes 210
python tools/noise_profile.py --config English/config.yaml --data_root D \
       --checkpoint English/checkpoints/en_model.pth.tar --speaker svarah_8a8e2655 \
       --gv_ratio Results/English/gv_ratio.json --gv_strength 1.0 --out Results/English/noise_profile.json
python English/inference.py  --data_root D --checkpoint English/checkpoints/en_model.pth.tar \
       --speaker svarah_8a8e2655 --gv_ratio Results/English/gv_ratio.json --gv_strength 1.0 \
       --prosody_fall --prosody_stress 1.0 --prosody_phrasing --duration_control 1.10 \
       --long_word_scale 1.25 --heavy_word_pause --denoise_profile Results/English/noise_profile.json \
       --edge_fade --out_dir Results/English
# gv_ratio.json: fastspeech2/postfilter_eval.py (Kaggle fs2-08-postfilter). Without the options: the model
alone.
```

Checkpoints: GitHub release v1.0 (`en_model.pth.tar`, `te_model.pth.tar`; SHA-256 in `<Lang>/checkpoints/README.md`). Inference alone needs only a checkpoint and the `stats.json` / `speakers.json` in `Results/<Lang>/`. **Kaggle jobs** (`kaggle/`, listed in `kaggle/README.md`) reproduce every stage, and each records the commit it ran. **Tests**: `pytest`.

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Appendix: deliverables checklist

Assignment item	Where
Two separate models, English (Svarah) and Telugu (IndicTTS Telugu), ~5 h train / ~1 h validation each	Sections 2.1, 3; `English/config.yaml`, `Telugu/config.yaml`
Data selection and subsampling strategy; train/validation separate with no overlap	Section 2.1; `fastspeech2/data/select_split.py`; `Evaluation/Data/accounting.json`
Dataset preparation and preprocessing, both languages	Section 2.2
Changes to the GitHub implementation for each language, particularly Telugu	Sections 1.2–1.3; `MODIFICATIONS.md` ("Telugu support at a glance", L1–L4, M1–M12, new files)
Speech for all test sentences, text preserved as provided	`Results/<Lang>/test_generated_audios/` (E1, E2a, E2b, E3, E4; T1, T2a, T2b, T3, T4); Section 5 (text check)
Folder structure: `English/`, `Telugu/` (train.py, preprocess.py, inference.py, config.yaml, checkpoints/), `Results/`, `Report/`, `README.md`, `requirements.txt`	repository root
Trained checkpoints, separately per language	GitHub release v1.0, linked from `<Lang>/checkpoints/README.md` and `README.md` (files exceed GitHub's 100 MB limit)

Training- and validation-loss plots, both models	`Results/<Lang>/training_loss.png`, `validation_loss.png`; Section 4
Variance-adaptor plots (duration, pitch, energy) for each test sentence, both languages	`Results/<Lang>/variance_adaptor_plots/{duration,pitch,energy}/`; Section 6
Report: implementation, architecture, modifications	Section 1
Report: data, subsampling, preprocessing, split, training configuration	Sections 2, 3
Report: loss curves, convergence, instability, over-fitting	Section 4
Report: generated speech per test sentence, pronunciation, quality	Sections 5.4, 5.5
Report: variance-adaptor analysis per test sentence	Section 6
Report: English vs Telugu, limitations, failures, improvements	Section 8
Report: reproducibility (dependencies, configs, commands)	Section 9; `README.md`